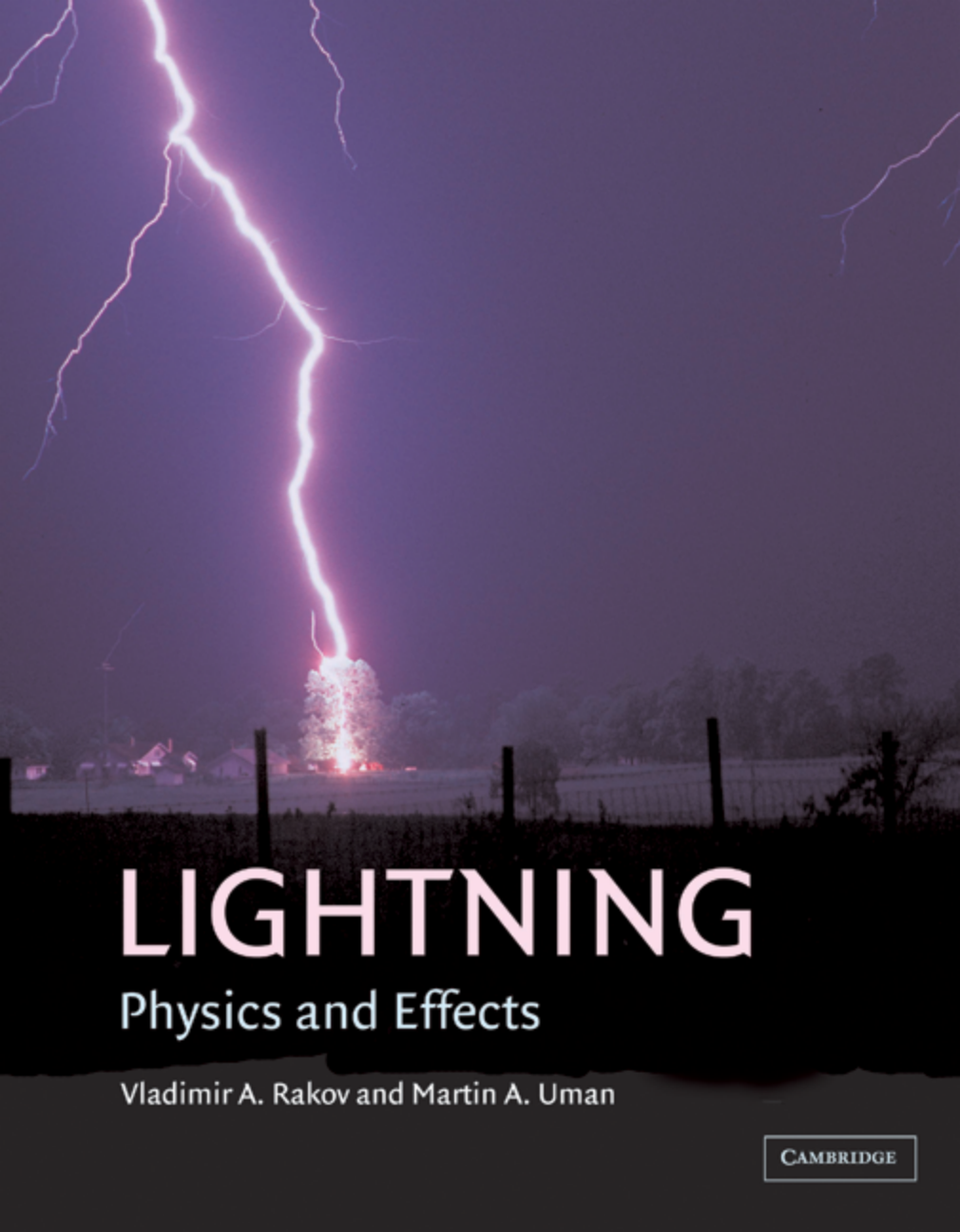




LIGHTNING

Physics and Effects

Vladimir A. Rakov and Martin A. Uman

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Lightning Physics and Effects

Lightning: Physics and Effects is the first book that covers essentially all aspects of lightning, including lightning physics, lightning protection, and the interaction of lightning with a variety of objects and systems as well as with the environment. It is written in a style that will be accessible to the technical non-expert and is addressed to anyone interested in lightning and its effects. This will include physicists, engineers working in the power industry and in the communications, computer, and aviation industries, meteorologists, atmospheric chemists, foresters, ecologists, physicians working in the area of electrical trauma, and architects. This comprehensive reference volume contains over 300 illustrations, 70 tables containing quantitative information, and over 6000 references and bibliography entries.

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Lightning

Physics and Effects

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Rakov's dedication

To my wife Lucy and our son Sergei

Uman's dedication

To my wife Dorit, our children Mara,
Jon, and Derek, and our grandchildren
Sara, Hunter, Hayden, Summer, and Isabella

Contents

<i>Preface</i>	<i>page x</i>	4.7. Subsequent leader	164
		4.8. Continuing current	173
1. Introduction	1	4.9. M-component	176
1.1. Historical overview	1	4.10. J- and K-processes	182
1.2. Types of lightning discharge and lightning terminology	4	4.11. Regular pulse bursts	188
1.3. Summary of salient lightning properties	6	4.12. Summary	190
1.4. The global electric circuit	6	<i>References and bibliography</i>	191
1.5. Regarding the utilization of lightning energy	12		
1.6. Summary	12	5. Positive and bipolar lightning discharges to ground	214
<i>References and bibliography</i>	12	5.1. Introduction	214
		5.2. Conditions conducive to the occurrence of positive lightning	217
2. Incidence of lightning	24	5.3. Characterization of positive lightning	218
2.1. Introduction	24	5.4. Bipolar lightning discharges to ground	232
2.2. Characterization of individual storms and storm systems	24	5.5. Summary	233
2.3. Thunderstorm days	35	<i>References and bibliography</i>	234
2.4. Thunderstorm hours	36		
2.5. Lightning flash density	37	6. Upward lightning initiated by ground-based objects	241
2.6. Long-term variations in lightning incidence	43	6.1. Introduction	241
2.7. Ratio of cloud flashes to cloud-to-ground flashes	44	6.2. General characterization	244
2.8. Characteristics of lightning as a function of season, location, and storm type	46	6.3. Overall electrical characteristics	247
2.9. Lightning incidence to various objects	49	6.4. Impulsive currents	250
2.10. Summary	52	6.5. Lightning current reflections within tall objects	252
<i>References and bibliography</i>	52	6.6. Electromagnetic fields due to lightning strikes to tall objects	259
		6.7. Acoustic output	260
3. Electrical structure of lightning-producing clouds	67	6.8. Summary	260
3.1. Introduction	67	<i>References and bibliography</i>	260
3.2. Cumulonimbus	68		
3.3. Non-cumulonimbus	91	7. Artificial initiation (triggering) of lightning by ground-based activity	265
3.4. Summary	93	7.1. Introduction	265
<i>References and bibliography</i>	93	7.2. Rocket-triggered lightning	266
		7.3. Other lightning triggering techniques	296
4. Downward negative lightning discharges to ground	108	7.4. Concluding remarks	299
4.1. Introduction	108	<i>References and bibliography</i>	299
4.2. General picture	108		
4.3. Initial breakdown	116	8. Winter lightning in Japan	308
4.4. Stepped leader	122	8.1. Introduction	308
4.5. Attachment process	137	8.2. Formation of winter thunderclouds	308
4.6. Return stroke	143	8.3. Evolution of winter thunderclouds	309

8.4. Characteristics of natural winter lightning	311	13.5. Whistlers	454
8.5. Rocket-triggered lightning in winter	314	13.6. Radio noise	459
8.6. Summary	316	13.7. Summary	461
<i>References and bibliography</i>	316	<i>References and bibliography</i>	461
9. Cloud discharges	321	14. Lightning effects in the middle and upper atmosphere	480
9.1. Introduction	321	14.1. Introduction	480
9.2. General information	322	14.2. Upward lightning channels from cloud tops	481
9.3. Phenomenology inferred from VHF–UHF imaging	326	14.3. Low-luminosity transient discharges in the mesosphere	482
9.4. Early (active) stage	329	14.4. Elves: low-luminosity transient phenomena in the lower ionosphere	492
9.5. Late (final) stage	338	14.5. Runaway electrons, X-rays, and gamma-rays	493
9.6. Comparison with ground discharges	340	14.6. Interaction of lightning and thundercloud electric fields with the ionosphere and the magnetosphere	495
9.7. Summary	341	14.7. Summary	497
<i>References and bibliography</i>	341	<i>References and bibliography</i>	497
10. Lightning and airborne vehicles	346	15. Lightning effects on the chemistry of the atmosphere	507
10.1. Introduction	346	15.1. Introduction	507
10.2. Statistics on lightning strikes to aircraft	348	15.2. Mechanism of NO production by return-stroke channels	511
10.3. Major airborne research programs	350	15.3. Laboratory determination of NO yield per unit energy	514
10.4. Mechanisms of lightning–aircraft interaction	353	15.4. Ground-based field determination of NO yield per lightning flash	514
10.5. Lightning test standards	362	15.5. Estimation of global NO production using the flash extrapolation approach (FEA)	516
10.6. Accidents	364	15.6. Estimation of NO production from airborne measurements	516
10.7. Summary	369	15.7. Estimation of NO production from extrapolation of nuclear explosion data	518
<i>References and bibliography</i>	369	15.8. Transport of lightning-produced trace gases	518
11. Thunder	374	15.9. Production of trace gases in the primitive Earth atmosphere and in the atmospheres of other planets	519
11.1. Introduction	374	15.10. Summary	520
11.2. Observations	374	<i>References and bibliography</i>	521
11.3. Generation mechanisms	377	16. Extraterrestrial lightning	528
11.4. Propagation	386	16.1. Introduction	528
11.5. Acoustic imaging of lightning channels	387	16.2. Detection techniques	530
11.6. Summary	389	16.3. Venus	531
<i>References and bibliography</i>	389	16.4. Jupiter	536
12. Modeling of lightning processes	394	16.5. Saturn	543
12.1. Introduction	394	16.6. Uranus	544
12.2. Return stroke	394	16.7. Neptune	545
12.3. Dart leader	415	16.8. Concluding remarks	546
12.4. Stepped leader	417	<i>References and bibliography</i>	547
12.5. M-component	419		
12.6. Other processes	420		
12.7. Summary	420		
<i>References and bibliography</i>	421		
13. The distant lightning electromagnetic environment: atmospherics, Schumann resonances, and whistlers	432		
13.1. Introduction	432		
13.2. Theoretical background	435		
13.3. Atmospherics (sferics)	443		
13.4. Schumann resonances	449		

17. Lightning locating systems	555	19. Lightning hazards to humans and animals	642
17.1. Introduction	555	19.1. Statistics	642
17.2. Electric and magnetic field amplitude techniques	556	19.2. Electrical aspects	644
17.3. Magnetic field direction finding	558	19.3. Medical aspects	646
17.4. Time-of-arrival technique	562	19.4. Personal safety	648
17.5. The US National Lightning Detection Network	565	19.5. Summary	648
17.6. Interferometry	568	<i>References and bibliography</i>	649
17.7. Ground-based optical direction finding	570	20. Ball lightning, bead lightning, and other unusual discharges	656
17.8. Detection from satellites	570	20.1. Introduction	656
17.9. Radar	572	20.2. Witness reports of ball lightning	658
17.10. Summary	573	20.3. Ball lightning statistics	662
<i>References and bibliography</i>	573	20.4. Ball lightning theories	662
18. Deleterious effects of lightning and protective techniques	588	20.5. Laboratory simulation of ball lightning	663
18.1. Introduction	588	20.6. Bead lightning	665
18.2. Basic mechanisms of lightning damage	588	20.7. Other types of unusual lightning and lightning-like discharges	666
18.3. Protection	589	20.8. Concluding comments	669
18.4. Lightning interaction with specific objects and systems	610	<i>References and bibliography</i>	669
18.5. Lightning test standards	624	<i>Appendix: books on lightning and related subjects</i>	675
18.6. Summary	626	<i>Index</i>	679
<i>References and bibliography</i>	626		

The plates will be found between pages 406 and 407

Preface

In this text, the first monograph to cover essentially all aspects of lightning, every effort has been made to present a balanced review of the present knowledge of lightning physics and lightning effects. The end-of-chapter reference and bibliography lists (a total of over 6000 entries) are essentially complete on most topics up to spring 2002. The Appendix contains a list of other books on lightning and related topics. Each of the authors has contributed to all chapters of the book; however the primary authorship by chapter is as follows. Chapters 1–9 and 12 were written by Rakov; Chapters 10, 13–15, and 18–20 by Uman; and Chapters 11, 16, and 17 were jointly written. General coordination of the work resulting in this book was conducted by Rakov.

Portions of the content of this book have been used for several years as the textbook for a one-semester senior- and graduate-level course on lightning at the University of Florida. The suggested content for a similar one-semester course would include Sections 1.2, 2.5, 2.9, subsections 3.2.1–3.2.7, Chapter 4, Section 7.2, and Chapters 12 and 17. The prerequisite for such a course would be an undergraduate course in electromagnetics or an undergraduate general physics course that covered electromagnetics in moderate detail.

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An electrically active thundercloud may be regarded as an electrostatic generator suspended in an atmosphere of low electrical conductivity. It is situated between two concentric conductors, namely, the surface of the earth and the electrosphere, the latter being the highly conducting layers of the atmosphere at altitudes above 50 to 60 km.

D.J. Malan (1967)

In this chapter, we introduce the basic lightning terminology (Section 1.2) and summarize the available quantitative information on various aspects of lightning in tabular form (Section 1.3). Additionally, we give a historical overview of the mythology and science of lightning covering the period from ancient times to the mid twentieth century (Section 1.1) and briefly discuss the global electric circuit that is generally thought to be energized by thunderstorms (Section 1.4). Finally, we consider whether the energy of lightning can be utilized and show that this is impractical (Section 1.5).

1.1. Historical overview

It is likely that lightning was present on Earth long before life evolved on our planet about three billion years ago. Further, it is possible that lightning played a role in producing the organic molecules necessary for the formation of every life form (Oparin 1938; Section 15.9 of this book). Harland and Hacker (1966) reported on a fossil glassy tube, referred to as a fulgurite, created by lightning 250 million years ago. Encounters of early humans with lightning undoubtedly were frightening and fascinating. All ancient civilizations incorporated lightning and thunder in their religious beliefs. Schonland (1964), Prinz (1977), Tomilin (1986), Wahlin (1986), Uman (1987, 2001), and Gary (1994) have reviewed mythological views of lightning in different cultures.

In ancient Egypt, the god Typhon (Seth) hurled the thunderbolts (lightning). A roll seal from Mesopotamia dated about 2200 BC shows a goddess standing on the shoulders of a winged creature and holding a bundle of thunderbolts in each hand (Prinz 1977, Fig. 1). Behind her, in a four-wheeled cart, is a weather god generating thunder with a whip. The similarly equipped weather god Teschup is seen on a Hittite relief found in northern Syria and dated about 900 BC. The thunderbolt is also the emblem of the

goddess Tien Mu in Chinese mythology. Tien Mu is among the five dignitaries of the “Ministry of Thunderstorms”, which is chaired by Lei Tsu, the God of Thunder, aided by Lei Kung, the drum-beating Count of Thunder. The ancient Vedic books of India describe how Indra, the son of Heaven and Earth, carried thunderbolts on his chariot. Many early statues of Buddha show him carrying in his right hand a thunderbolt with prongs at each end.

According to Prinz (1977), around 700 BC the ancient Greeks began using the lightning symbols of the Middle East in their art, attributing them primarily to Zeus, their supreme god. A lightning flash was one of the chief signs of the displeasure of Zeus in ancient Greece and of Jupiter in ancient Rome. In Rome, the laurel bush was considered, according to Pliny, to be immune from lightning. For this reason, the emperor Tiberius wore a wreath of laurel during thunderstorms. A very powerful political body, the College of Augurs, was formed about 300 BC to determine the views of Jupiter regarding Roman State affairs. This task was accomplished by making observations of three classes of objects in the sky: birds, meteors, and lightning.

In ancient Scandinavia, lightning was believed to be produced by the magic hammer Mjollnir of the god Thor, who hurled it from a chariot rolling thunderously upon the clouds. The Buryats, living in the area of Lake Baikal (Russia), believed that their god produced lightning by throwing stones from the sky. Some Indian tribes of North America, as well as certain tribes in southern Africa, hold the belief that lightning is produced by a magical thunderbird, which dives from the clouds to earth.

There exists a long record of lightning damage to tall structures, particularly churches, covering the period from the Middle Ages to the modern era. For example, the Campanile of St. Mark in Venice, which is about 100 m high, was damaged or destroyed by lightning in 1388, 1417, 1489, 1548, 1565, 1653, 1745, 1761, and 1762. In 1766, a

lightning protective system, invented in 1752 by Benjamin Franklin and often referred to as a Franklin rod system, was installed and no further lightning damage has occurred since. In 1718, 24 church towers along the Brittany coast of France were damaged by lightning, apparently during the same storm. In 1769, the Church of St. Nazaire in Brescia, Italy, was struck by lightning. About 100 tons of gunpowder had been placed in the vaults of the church for safekeeping, and when the lightning discharge caused it to explode, about one-sixth of the city was destroyed, more than 3000 people being killed.

There are, however, many historic buildings which have never been seriously damaged by lightning, apparently because they had, in effect, a lightning protective system equivalent to that proposed later by Franklin. For example, the Temple in Jerusalem, originally built by Solomon, experienced no apparent damage from lightning over a period of a thousand years. Other examples are the 62 m high monument erected in 1677 in commemoration of the Great Fire of London and the cathedral of Geneva, the most prominent building in that city. This cathedral was immune from lightning damage for 300 years prior to being equipped with a Franklin rod system. In 1773, Franklin pointed out that “buildings that have their roofs covered with lead or other metal, and spouts of metal continued from the roof into the ground are never hurt by lightning; as whenever it falls on such a building, it passes in the metal and not in the walls” (Schonland 1964, p. 14). More information on the lightning protection of buildings, including historical aspects, is found in [Section 18.3](#).

The practice of ringing church bells during thunderstorms in an attempt to disperse lightning existed for many centuries in Europe. Since bell towers are usually preferred targets for lightning because of their relatively large height (see [Section 2.9](#)), this practice caused the deaths of many of those pulling the ropes. In fact, over a period of 33 years, lightning struck 386 church towers and killed 103 bell-ringers, as reported in a book by Fischer published in Munich in 1784 (Schonland 1964, p. 9).

Many ships with wooden masts have been severely damaged or totally destroyed by lightning, as discussed in [subsection 18.4.1](#). Harris (1834, 1838, 1839, 1843) reported that from 1799 to 1815 there were 150 cases of lightning damage to British naval vessels. One ship in eight was set on fire, nearly 100 lower masts were destroyed, about 70 sailors were killed, and more than 130 people were wounded. In 1798, the 44-gun ship *Resistance* exploded as a result of a lightning discharge.

Systematic studies of thunderstorm electricity can be traced back to 10 May 1752 in the village of Marly-la-Ville, near Paris. On that day, in the presence of a nearby storm, a retired French dragoon, acting on instructions from Thomas-François Dalibard, drew sparks from a tall

iron rod that was insulated from ground by wine bottles. The results of this experiment, proposed by Benjamin Franklin, provided the first direct proof that thunderclouds contain electricity, although several scientists had previously noted the similarity between laboratory sparks and lightning (Prinz 1977; Tomilin 1986). The Marly experiment was repeated thereafter in several countries including Italy, Germany, Russia, Holland, England, Sweden, and again France. Franklin himself drew sparks from the probably moist hemp string of a kite after the success at Marly, but before he knew about this (Cohen 1990). Not only kites but also balloons, mortars, and rockets were used to extend conducting strings into the electric field of the cloud (Prinz 1977, Fig. 5). In all these experiments, the metallic rod (such as in the experiment at Marly) or the conducting string was polarized by the electric field of the cloud, so that charges of opposite polarities accumulated at the opposite ends of the conductor. As the gap between the bottom end of the conductor and ground was decreased, a spark discharge to ground occurred. The scale and effect of this spark discharge are orders of magnitude smaller than those of lightning. In designing his experiments, Franklin did not consider the possibility of a direct lightning strike to the rod or the kite. Such a strike would almost certainly have killed the experimenter. Thus all those who performed these experiments risked their lives, but there is only one case on record in which a direct strike did occur in such experiments. This happened on 6 August 1753 in St. Petersburg, Russia when Georg Richmann, who had previously done Franklin’s experiment, was killed by a direct lightning strike to an ungrounded rod. Interestingly, Richmann was not in contact with the rod, and what caused his death appeared to be a ball lightning that came out of the rod and went to his forehead. This accident is discussed further in [Section 20.1](#).

Franklin also showed that lightning flashes originate in clouds that are “most commonly in a negative state of electricity, but sometimes in a positive state” (Franklin 1774). Even before the experiment at Marly, Franklin had proposed the use of grounded rods for lightning protection. Originally, he thought that the lightning rod would silently discharge a thundercloud and thereby would prevent the initiation of lightning. Later, Franklin stated that the lightning rod had a dual purpose: if it cannot prevent the occurrence of lightning, it offers a preferred attachment point for lightning and then a safe path for the lightning current to ground. It is in the latter manner that lightning rods, often referred to as Franklin rods, actually work ([Section 18.3](#)), as suggested by Lomonosov in 1753 (Tomilin 1986); a lightning rod cannot appreciably alter the charge in a cloud. One convincing demonstration of the effectiveness of Franklin rods took place in Siena, Italy, on 18 April 1777. On that day, a large number of people gathered near the 102 m tower of the city hall, which had been repeatedly struck and

damaged by lightning prior to the installation of lightning rods in 1777. A thunderstorm rumbled into the area, and the crowd saw lightning strike the lightning protective system without damaging the tower. More information on the contributions of Benjamin Franklin to the study of lightning and lightning protection is found, for example, in Dibner (1977) and Krider (1996b).

In 1876, James Clerk Maxwell suggested that Franklin rod systems attracted more lightning strikes than the surrounding area. He proposed that a gunpowder building be completely enclosed with metal of sufficient thickness, forming what is now referred to as a Faraday cage. If lightning were to strike a metal-enclosed building, the current would be constrained to the exterior of the metal enclosure, and it would not even be necessary to ground this enclosure. In the latter case, the lightning would merely produce an arc from the enclosure to earth. The Faraday cage effect is provided by all-metal cars and airplanes. Modern steel-frame buildings with reinforcing metal bars in the concrete foundation connected to the building steel provide a good approximation to a Faraday cage. As the spacing between conductors increases, however, the efficiency of the lightning protection decreases. In practice, a combination of the Franklin rod system concept and the Faraday cage concept is often used. Modern lightning protection schemes for structures containing computers or other sensitive electronics employ a technique known as topological shielding with surge suppression (subsection 18.3.6), which can be viewed as a generalization of the Faraday cage concept.

In the following, we briefly review the history of lightning research from the latter part of the nineteenth century to the middle of the twentieth century. In the late nineteenth century, photography and spectroscopy became available as diagnostic tools for lightning research. Among the early investigators of the lightning spectrum were Herschel (1868), Gibbons (1871), Holden (1872), and Clark (1874). It was Herschel who first identified a nitrogen line as being the brightest in the visible spectrum and who noted that the relative intensities of the lines change from spectrum to spectrum. Schuster (1880) made the first systematic identification of the lines in the spectrum of lightning. Dufay (1949) and Israel and Fries (1956) were the first to consider the spectrum of lightning as a source of quantitative information about the physical conditions in and around the lightning channel. Slipher (1917) obtained the first photographic record of the spectrum of lightning and noted that there were both line and continuum emissions. Spectroscopic studies of lightning are reviewed by Uman (1969, 1984, Chapter 5) and Orville (1977).

Among the early investigators who used stationary or moving photographic cameras were Hoffert (1889), Weber (1889), Walter (1902, 1903, 1910, 1912, 1918), and Larsen (1905). Time-resolved photographs showing that

lightning flashes often contain two or more strokes, similar to that shown in Fig. 4.1a, were obtained. The invention of the streak camera (Boys 1926) and its further improvement (Boys 1929; Malan 1950, 1957) facilitated the major advances in lightning research made by B.F.J. Schonland, D.J. Malan, and their co-workers in South Africa during the 1930s. For example, it was shown conclusively by the South African researchers that lightning strokes lowering negative charge to ground are composed of a downward leader and an upward return stroke and that the first-stroke leader is stepped (see Fig. 4.2). Schonland (1956) summarized the main results of the South African studies. (Much of the presently used lightning terminology was introduced by the South African researchers.) The results obtained in South Africa have been confirmed and extended by investigators using streak cameras in the United States, Russia, France, Japan, and Switzerland. A review of photographic studies of lightning through the 1960s is found in Uman (1969, 1984, Chapter 2).

The first estimates of lightning peak current, inferred from the residual magnetization of pieces of basalt placed near the strike object, were made by Pockels (1900). Further information on the estimates of lightning currents obtained using magnetizable materials is found in Uman (1969, 1984, Chapter 4). The first oscillographic recordings of lightning current waveforms were obtained using tethered balloons in Russia (Stekolnikov and Valeev 1937) and in England (Davis and Standring 1947), and on the Empire State Building in New York City (McEachron 1939, 1941; Hagenguth and Anderson 1952). The most comprehensive data on lightning current waveforms to date were acquired by K. Berger and his associates on two instrumented towers on Monte San Salvatore in Switzerland, as discussed in Chapters 4, 5, and 6.

C.T.R. Wilson, who received a Nobel Prize for his invention of the cloud chamber to track high-energy particles, was the first to use electrostatic field measurements to infer the charge structure of thunderclouds (subsection 3.2.2) as well as the charges involved in the lightning discharge. Simpson and Scrase (1937) and Simpson and Robinson (1941) made the earliest measurements of electric fields inside thunderclouds and used these measurements to infer cloud charge structure (subsection 3.2.3). The first multiple-station measurements of the electromagnetic fields on ground from relatively close lightning were performed by Workman *et al.* (1942) and Reynolds and Neill (1955). Early measurements of the electric fields of distant lightning in the frequency range from a few to a few tens of kilohertz, called atmospherics, are discussed in Section 13.3. Austin (1926), Appleton *et al.* (1926), Wattson-Watt (1929), Norinder (1936), and Chapman (1939) were among the first to study atmospherics. Norinder and Dahle (1945) made an attempt to relate lightning magnetic field measurements to

the current in the lightning channel. The modern era of electric and magnetic field measurements relating to lightning can be traced to the 1970s, when the first field records on microsecond and submicrosecond time scales were reported (see [Chapters 4 and 9](#)). More information on the early measurements of the electric and magnetic fields due to lightning is found in Uman (1969, 1984, [Chapter 3](#)).

1.2. Types of lightning discharge and lightning terminology

Lightning, or the lightning discharge, in its entirety, whether it strikes ground or not, is usually termed a “lightning flash” or just a “flash”. A lightning discharge that involves an object on ground or in the atmosphere is sometimes referred to as a “lightning strike”. A commonly used non-technical term for a lightning discharge is “lightning bolt”. The terms “stroke” or “component stroke” apply only to components of cloud-to-ground discharges. Each stroke involves a downward leader and an upward return stroke and may involve a relatively low level “continuing current” that immediately follows the return stroke. Transient processes occurring in a lightning channel while it carries continuing current are termed M-components. First strokes are initiated by “stepped” leaders while subsequent strokes following previously formed channels are initiated by “dart” or “dart-stepped” leaders.

From the observed polarity of the charge effectively lowered to ground and the direction of propagation of the initial leader, four different types of lightning discharges between cloud and Earth have been identified. The term “effectively” found here and elsewhere in this book in a similar context (although often it is omitted for simplicity) is used to indicate that individual charges are not transported all the way from the cloud to ground during the lightning processes. Rather, the flow of electrons (the primary charge carriers) in one part of the lightning channel results in the flow of other electrons in other parts of the channel, as discussed by Uman (1987, 2001). For example, individual electrons in the lightning channel move only a few meters during a return stroke that transfers a coulomb or more of charge to ground.

The four types of lightning, illustrated in [Fig. 1.1](#), are (a) downward negative lightning, (b) upward negative lightning, (c) downward positive lightning, and (d) upward positive lightning. Discharges of all four types can be viewed as effectively transporting cloud charge to the ground and therefore are usually termed cloud-to-ground discharges (sometimes referred to as CGs). It is believed that downward negative lightning flashes, type (a), account for about 90 percent or more of global cloud-to-ground lightning, and that 10 percent or less of cloud-to-ground dis-

charges are downward positive lightning flashes (type (c)). Upward lightning discharges, types (b) and (d), are thought to occur only from tall objects (higher than 100 m or so) or from objects of moderate height located on mountain tops ([Chapter 6](#)). Rocket-triggered lightning, discussed in [Chapter 7](#), is similar in its phenomenology to the upward lightning initiated from tall objects. The term “initial continuous current,” as differentiated from continuing current, is used to denote the relatively low-level current flowing during the initial stage of upward ([Chapter 6](#)) and rocket-triggered ([Chapter 7](#)) lightning. The downward negative lightning discharge is considered in [Chapter 4](#) and the positive lightning discharge in [Chapter 5](#). Additionally discussed in [Chapter 5](#) are lightning flashes that transfer both negative and positive charges to ground. The majority of lightning discharges, probably three-quarters, do not involve ground. These are termed cloud discharges and sometimes are referred to as ICs. Cloud discharges include intracloud, intercloud, and cloud-to-air discharges and are considered in [Chapter 9](#). Unusual forms of lightning and lightning-like discharges (including ball lightning) are discussed in [Chapter 20](#).

There are three possible modes of charge transfer to ground in lightning discharges. It is convenient to illustrate these for the case of negative subsequent strokes. In negative subsequent strokes these three modes are represented by (a) dart-leader–return-stroke sequences, (b) continuing currents, and (c) M-components. [Figure 1.2](#) schematically shows current profiles corresponding to these three modes, which we now discuss.

(a) In a leader–return-stroke sequence, the descending leader creates a conductive path between the cloud charge source and ground and deposits negative charge along this path. The following return stroke traverses that path, moving from ground toward the cloud charge source, and neutralizes the negative leader charge. Thus, both leader and return-stroke processes serve to transport effectively negative charge from the cloud to ground.

(b) The lightning continuing current can be viewed as a quasi-stationary arc between the cloud charge source and ground. The typical arc current is tens to hundreds of amperes, and the duration is up to some hundreds of milliseconds.

(c) Lightning M-components can be viewed as perturbations (or surges) in the continuing current and in the associated channel luminosity. It appears that an M-component involves the superposition of two waves propagating in opposite directions (see [Fig. 1.2](#)). The spatial front length for M-component waves is of the order of a kilometer (shown shorter in relation to the cloud height in [Fig. 1.2](#), for illustrative purposes), while for dart-leader and return-stroke waves the spatial front lengths are of the

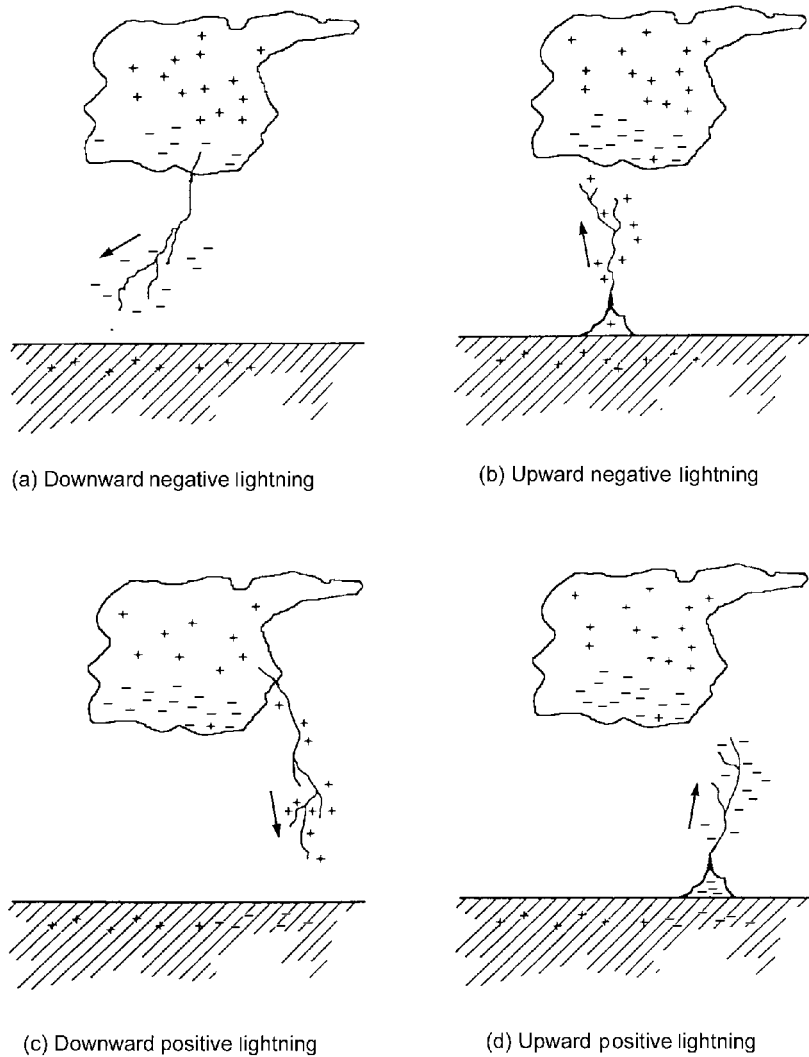


Fig. 1.1. Four types of lightning effectively lowering cloud charge to ground. Only the initial leader is shown for each type. In each lightning-type name given below the sketch, the direction of propagation of the initial leader and the polarity of the cloud charge effectively lowered to ground are indicated.

order of 10 and 100 m, respectively. The M-component mode of charge transfer to ground requires the existence of a grounded channel carrying a continuing current that acts as a wave-guiding structure. In contrast, the leader–return-stroke mode of charge transfer to ground occurs only in the absence of such a conducting path to ground. In this latter mode, the wave-guiding structure is not available and is created by the leader. For all the processes shown in Fig. 1.2, the channel conductivity is of the order of 10^4 S m^{-1} , except for the channel section between the dart-leader tip and ground shown by a broken line. For this latter channel section, the conductivity is about 0.02 S m^{-1} (Rakov 1998). Thus, the primary distinction between the leader–return-stroke and M-component modes is the availability of a conducting

path to ground. It is possible that, as the conductivity of the path to ground decreases, the downward M-component wave can transform to a dart leader.

We now define the terms “leader” and “streamer”, as they are used in this book. Any self-propagating electrical discharge creating a channel with electrical conductivity of the order 10^4 S m^{-1} (comparable to that of carbon) is called a leader. Streamers, on the other hand, are characterized by much lower electrical conductivity; the air behind the streamer tip remains essentially an insulator (e.g., Bazelyan *et al.* 1978). A corona or point discharge consists of numerous individual streamers. Corona discharge is confined to the immediate vicinity of an “electrode” such as a grounded object, a leader tip, the lateral surface of the leader channel,

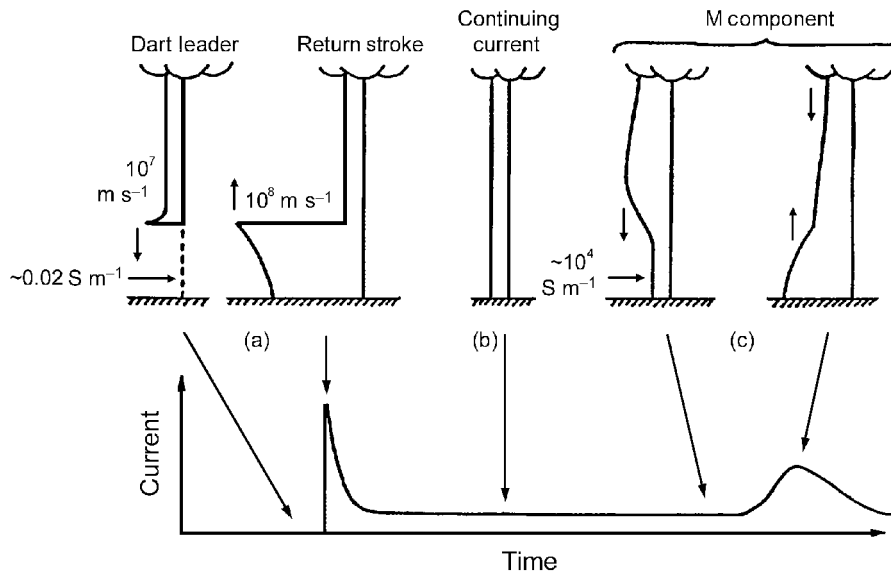


Fig. 1.2. Schematic representation of current versus height profiles for three modes of charge transfer to ground in negative lightning subsequent strokes: (a) dart-leader–return-stroke sequence, (b) continuing current, and (c) M-component. The corresponding current versus time waveform represents the current at the ground.

or a hydrometeor, that is, it is not a self-propagating discharge. It is worth noting that the terms leader and streamer in the lightning literature are sometimes used interchangeably, the term streamer in most cases being used to denote a low-luminosity leader, particularly the upward connecting leader (Section 4.5).

1.3. Summary of salient lightning properties

The salient properties of downward negative lightning discharges, the most common type of cloud-to-ground lightning, are summarized in Table 1.1. Negative lightning is discussed in detail in Chapter 4. Properties of positive and bipolar lightning are discussed in Chapter 5 and those of cloud discharges in Chapter 9. Characteristics of microsecond-scale electric field pulses associated with various lightning processes are summarized in Table 1.2.

Various lightning processes emit electromagnetic signals with a peak in the radio-frequency spectrum at 5 to 10 kHz when observed at distances beyond 50 km or so. At frequencies higher than that of the spectrum peak, the spectral amplitude is approximately inversely proportional to the frequency up to 10 MHz or so and inversely proportional to the square root of frequency from about 10 MHz to 10 GHz (Cianos *et al.* 1973). The mechanisms of radiation in the high-frequency (HF) region of the spectrum, 3–30 MHz, and above are not fully understood. It is thought that this radiation is caused by numerous small sparks occurring during the formation of new channels,

that is, by the electrical breakdown of air rather than by high-current pulses propagating in pre-existing channels.

1.4. The global electric circuit

Shortly after the experiment at Marly that confirmed Franklin's conjecture regarding the electrical nature of thunderstorms (Section 1.1), Lemonnier (1752) discovered atmospheric electrical effects in fair weather. Further research established that the Earth's surface is charged negatively and the air is charged positively, the associated vertical electric field in fair weather being about 100 V m^{-1} near the Earth's surface.

1.4.1. Conductivity of the atmosphere

The atmosphere below about 50 km is conducting, owing to the presence of ions created by both cosmic rays and the natural radioactivity of the Earth. Small ions, those with diameters of 0.1 to 1 nm and lifetimes of about 100 s, are the primary contributors to the conductivity of the lower atmosphere. Free electrons at these heights are attached to neutrals on time scales of the order of microseconds, and their contribution to the conductivity of the atmosphere below about 50 km can be neglected (Gringel *et al.* 1986; Reid 1986). Above 60 km or so, free electrons become the major contributors to the atmospheric conductivity. The average production rate of ions at sea level is one to 10 million pairs per cubic meter per second. Cosmic rays and natural radioactivity contribute about equally to the production of ions at the land surface. Since large water surfaces have no

Table 1.1. *Characterization of negative cloud-to-ground lightning*

Parameter	Typical value ^a
Stepped leader	
Step length, m	50
Time interval between steps, μs	20–50
Step current, kA	> 1
Step charge, mC	> 1
Average propagation speed, m s^{-1}	2×10^5
Overall duration, ms	35
Average current, A	100–200
Total charge, C	5
Electric potential, MV	~ 50
Channel temperature, K	$\sim 10\,000$
First return stroke ^b	
Peak current, kA	30
Maximum current rate of rise, $\text{kA } \mu\text{s}^{-1}$	$\geq 10\text{--}20$
Current risetime (10–90 percent), μs	5
Current duration to half-peak value, μs	70–80
Charge transfer, C	5
Propagation speed, m s^{-1}	$(1\text{--}2) \times 10^8$
Channel radius, cm	$\sim 1\text{--}2$
Channel temperature, K	$\sim 30\,000$
Dart leader	
Speed, m s^{-1}	$(1\text{--}2) \times 10^7$
Duration, ms	1–2
Charge, C	1
Current, kA	1
Electric potential, MV	~ 15
Channel temperature, K	$\sim 20\,000$
Dart-stepped leader	
Step length, m	10
Time interval between steps, μs	5–10
Average propagation speed, m s^{-1}	$(1\text{--}2) \times 10^6$
Subsequent return stroke ^b	
Peak current, kA	10–15
Maximum current rate of rise, $\text{kA } \mu\text{s}^{-1}$	100
10–90 percent current rate of rise, $\text{kA } \mu\text{s}^{-1}$	30–50
Current risetime (10–90 percent), μs	0.3–0.6
Current duration to half-peak value, μs	30–40
Charge transfer, C	1
Propagation speed, m s^{-1}	$(1\text{--}2) \times 10^8$
Channel radius, cm	$\sim 1\text{--}2$
Channel temperature, K	$\sim 30\,000$
Continuing current (longer than $\sim 40\text{ ms}$) ^c	
Magnitude, A	100–200
Duration, ms	~ 100
Charge transfer, C	10–20
M-component ^b	
Peak current, A	100–200
Current risetime (10–90 percent), μs	300–500
Charge transfer, C	0.1–0.2

Table 1.1. (*cont.*)

Parameter	Typical value ^a
Overall flash	
Duration, ms	200–300
Number of strokes per flash ^d	3–5
Interstroke interval, ms	60
Charge transfer, C	20
Energy, J	$10^9\text{--}10^{10}$

^a Typical values are based on a comprehensive literature search and unpublished experimental data acquired by the University of Florida Lightning Research Group.

^b All current characteristics for return strokes and M-components are based on measurements at the lightning channel base.

^c About 30 to 50 percent of lightning flashes contain continuing currents having durations longer than $\sim 40\text{ ms}$.

^d About 15 to 20 percent of lightning flashes are composed of a single stroke.

significant radioactive emanation, the production of ions over oceans is about one-half of that over land. At altitudes of roughly 1 km and greater, cosmic rays are responsible for most of the ions in the fair weather atmosphere, regardless of the presence of land below. The ionization rate depends on magnetic latitude and on the solar activity.

The electrical conductivity of the air at sea level is about 10^{-14} S m^{-1} , and it increases rapidly with altitude. A diagram illustrating conductivity variations up to an altitude of 120 km is shown in Fig. 1.3. The various regions of the atmosphere are named in Fig. 14.1, and typical values of electron density and collision frequency as a function of height are found in Figs. 13.5 and 13.6, respectively.

At a height of 35 km, where the air density is about one percent of that at the Earth's surface, the electrical conductivity is greater than 10^{-11} S m^{-1} , which is more than three orders of magnitude higher than at sea level. For comparison, the average electrical conductivity of the Earth is about 10^{-3} S m^{-1} . As seen in Fig. 1.3, a considerable variation of conductivity exists at the same altitude for different measurements, about six orders of magnitude at 60 km. Above about 80 km, the conductivity becomes anisotropic because of the influence of the geomagnetic field (Section 13.2), and there are diurnal variations due to solar photoionization processes.

Blakeslee *et al.* (1989) reported, from high-altitude U-2 airplane measurements, that the conductivity near 20 km was relatively steady above storms, variations being less than ± 15 percent. However, a number of balloon measurements of the electrical conductivity between 26 and 32 km over thunderstorms suggest that some of the time the storm may significantly (up to a factor 2) perturb the

Table 1.2. *Characterization of microsecond-scale electric field pulses associated with various lightning processes. Adapted from Rakov et al. (1996)*

Type of pulses	Dominant polarity ^a		Typical ^b total pulse duration, μs	Typical ^b time interval between pulses, μs	Comments
	Atmospheric electricity sign convention	Physics sign convention			
Return stroke in negative ground flashes	positive	negative	30–90 (zero-crossing time)	60×10^3	3–5 pulses per flash
Stepped leader in negative ground flashes	positive	negative	1–2	15–25	Within 200 μs just prior to a return stroke
Dart-stepped leader in negative ground flashes	positive	negative	1–2	6–8	Within 200 μs just prior to a return stroke
Initial breakdown in negative ground flashes	positive	negative	20–40	70–130	Some milliseconds to some tens of milliseconds before the first return stroke
Initial breakdown in cloud flashes	negative	positive	50–80	600–800	The largest pulses in a flash
Regular pulse burst in both cloud and negative ground flashes	Both polarities are about equally probable		1–2	5–7	Occur later in a flash; 20–40 pulses per burst
Narrow bipolar pulses	negative	positive	10–20	-	Probably associated with the initial breakdown in cloud flashes

^a The polarity of the initial half cycle in the case of bipolar pulses.

^b Typical values are based on a comprehensive literature search and unpublished experimental data acquired by the University of Florida Lightning Research Group.

conductivity (Bering *et al.* 1980b; Holzworth *et al.* 1986; Pinto *et al.* 1988; Hu *et al.* 1989). It is usually assumed that the atmosphere above a height of 60 km or so, under quasi-static conditions, becomes conductive enough to consider it an equipotential region. The electrical conductivity increases abruptly above about 60 km because of the presence of free electrons (Roble and Tzur 1986; Reid 1986). This region of atmosphere just above 60 km or so where free electrons are the major contributors to the conductivity is sometimes referred to as the electrosphere (e.g., Chalmers 1967) or “equalizing” layer (Dolezalek 1972). At 100 km altitude (in the lower ionosphere) the conductivity is about 12 ± 2 orders of magnitude (depending on the local time of the day) greater than the conductivity near the Earth’s surface, that is, the conductivity at 100 km is comparable to the conductivity of the Earth, whether land or sea (Rycroft 1994).

1.4.2. Fair-weather electric field

As noted above, the electric field near the Earth’s surface under fair-weather (also called fine-weather) conditions is about 100 V m^{-1} . The electric field vector is directed downward. This downward-directed field is defined as positive according to the “atmospheric electricity” sign convention. According to the alternative sign convention, sometimes referred to as the “physics” sign convention, a downward directed electric field is negative because it is in the direction opposite to that of the radial coordinate vector of the spherical coordinate system whose origin is at the Earth’s center. We will use the physics sign convention in this chapter (both sign conventions are given in Table 1.2) and in Chapters 3 and 8. However, in order to minimize conflict with the existing literature, we will use the atmospheric electricity sign convention in Chapters 4, 5, 6, 7, and 12, and we will employ both the atmospheric electricity and physics

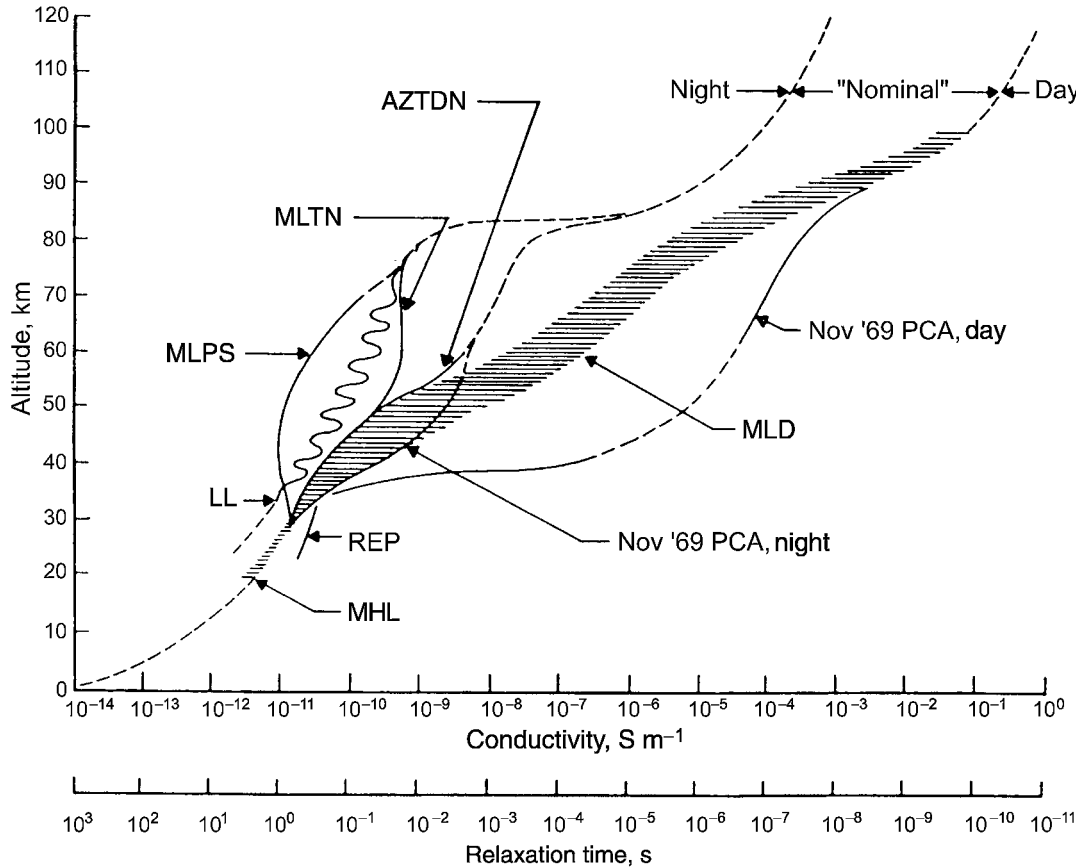


Fig. 1.3. Electrical conductivity σ and corresponding relaxation time $\tau = \epsilon_0 \sigma^{-1}$, where $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$, versus altitude under a variety of geophysical conditions. LL, low latitude, “wavy”; MLPS, mid-latitude pre-sunrise (unusual); MLTN, mid-latitude typical night (high-latitude, quiet); AZTDN, auroral zone, typical disturbed night; MLD, mid-latitude day, quiet; MHL, mid-high-latitude, typical of ~ 100 measurements; REP, relativistic electron (energy from a few MeV to 10 MeV) precipitation event (unusual); PCA, polar cap absorption event (an unusually large flux of energetic, ~ 100 MeV solar protons within the polar cap). Adapted from Hale (1984).

sign conventions in Chapter 9. We will indicate explicitly the direction of the electric field vector when appropriate. The magnitude of the fair-weather electric field decreases with increasing altitude. For example, according to Volland (1984),

$$E(z) = -[93.8 \exp(-4.527z) + 44.4 \exp(-0.375z) + 11.8 \exp(-0.121z)] \quad (1.1)$$

where $E(z)$ is the electric field in V m^{-1} (the negative sign indicates that the electric field vector is directed downward) and z is the altitude in kilometers. This equation is valid at mid-latitudes below about 60 km altitude and outside thunderstorms or cloudy areas. According to Eq. 1.1, the electric field at the ground is 150 V m^{-1} and at 10 km altitude decreases to about 3 percent of its value at the ground. The electric field magnitude normally drops to around 300 mV m^{-1} at 30 km at mid-latitudes (Gringel *et al.* 1986) and to $1 \mu\text{V m}^{-1}$ or so at about 85 km (Reid 1986).

1.4.3. “Classical” view of atmospheric electricity

The evaluation of the line integral of the electric field intensity from the Earth’s surface to the height of the electrosphere yields the negative of the potential of the electrosphere, sometimes termed the ionospheric potential (e.g., Markson 1976), with respect to Earth potential. The potential of the electrosphere is positive with respect to the Earth and its magnitude is about 300 kV, most of the voltage drop taking place below 20 km where the electric field is relatively large. The overall situation is often visualized as a lossy spherical capacitor (e.g., Uman 1974), the outer and inner shells of which are the electrosphere and Earth’s surface, respectively. According to this model, the Earth’s surface is negatively charged, the total charge magnitude being roughly $5 \times 10^5 \text{ C}$, while an equal positive charge is distributed throughout the atmosphere. Little charge resides on the electrosphere “shell”. Further, most of the net positive charge is found within 1 km of the Earth’s

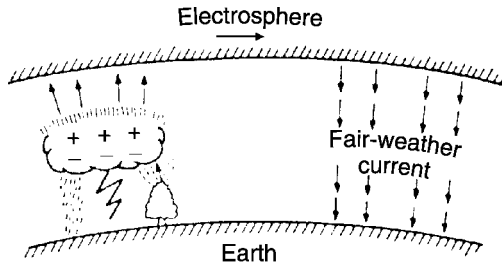


Fig. 1.4. Illustration of the global electric circuit. Shown schematically under the thundercloud are precipitation, lightning, and corona. Adapted from Pierce (1974).

surface, and more than 90 percent of this charge within 5 km (MacGorman and Rust 1998). Because the atmosphere between the capacitor “shells” is weakly conducting, there is a fair-weather leakage current of the order of 1 kA (2 pA m^{-2} ; $1 \text{ pA} = 10^{-12} \text{ A}$) between the shells that would neutralize the charge on the Earth and in the atmosphere on a time scale of roughly 10 minutes (depending on the amount of pollution) if there were no charging mechanism to replenish the neutralized charge. Since the capacitor is observed to remain charged, there must be a mechanism or mechanisms acting to resupply that charge. Wilson (1920) suggested that the negative charge on the Earth is maintained by the action of thunderstorms. Thus all the stormy-weather regions worldwide (on average, at any time a total of about 2000 thunderstorms are occurring, over about 10 percent of the Earth’s surface) constitute the global thunderstorm generator, while the fair-weather regions (about 90 percent of the globe) can be viewed as a resistive load. Lateral currents are assumed to flow freely along the highly conducting Earth’s surface and in the electrosphere. The fair-weather current, of the order of 1 kA, must be balanced by the total generator current, which is composed of currents associated with corona, precipitation, and lightning discharges. The total current flowing from cloud tops to the electrosphere is, on average, about 0.5 A per thunderstorm (Gish and Wait 1950). The global electric circuit concept is illustrated in Fig. 1.4. Negative charge is brought to Earth mainly by lightning discharges (most of which transport negative charge to ground) and by corona current under thunderclouds. The net precipitation current is thought to transport positive charge to ground, and its magnitude is comparable to the lightning current (Wahlin 1986). Positive charge is presumed to leak from cloud tops to the electrosphere. If we divide the potential of the electrosphere, 300 kV, by the fair-weather current, 1 kA, the effective load resistance is 300Ω .

The diurnal variation of the fair-weather field as a function of universal time over the oceans, the so-called Carnegie curve, named after the research vessel *Carnegie* on which the measurements were made (Torreson *et al.* 1946), appears to follow the diurnal variation of the total

worldwide thunderstorm area (Whipple and Scrase 1936). Both characteristics exhibit maximum values near 1900 UT and minimum values near 0400 UT. However, the annual variation of the fair-weather electric field is not in phase with the annual variation of thunderstorm activity throughout the world (Imyanitov and Chubarina 1967). Füllekrug *et al.* (1999) found that the hourly contribution of global cloud-to-ground lightning activity (as represented by magnetic field measurements in the frequency range 10–135 Hz) to the fair-weather electric field in the Antarctic during December 1992 was about 40 ± 10 percent, and that the contribution to hourly departures from the mean diurnal variation of the electric field was about 25 ± 10 percent. Holzworth *et al.* (1984) showed that significant time variations could occur in the global fair-weather current on time scales of 10 minutes to several hours.

According to the classical picture of atmospheric electricity, the layer of the atmosphere extending from about 15 to 200 km and including the stratosphere, mesosphere, and the lower portion of the thermosphere (Chapter 15 and Fig. 14.1), should be “passive”. However, some rocket measurements indicate the existence of strong electric fields (in the volts per meter range, which is orders of magnitude higher than expected in the mesosphere, at altitudes of 50–85 km) of unknown origin (Bragin *et al.* 1974; Tyutin 1976; Hale and Croskey 1979; Hale *et al.* 1981; Maynard *et al.* 1981; Gonzalez *et al.* 1982). Interestingly, these abnormally strong electric fields are observed near the 60 to 65 km height region, where the “equalizing” layer (the electrosphere) is presumed to exist. Their origin remains a subject of controversy. Any plausible explanation of this phenomenon must involve either a local mesospheric field-generation mechanism or a dramatic local decrease in conductivity.

1.4.4. Maxwell current density

The Maxwell current density $\mathbf{J}_M$ associated with a thunderstorm is defined as the sum of four terms (e.g., Krider and Musser 1982):

$$\mathbf{J}_M = \mathbf{J}_E + \mathbf{J}_C + \mathbf{J}_L + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (1.2)$$

where $\mathbf{J}_E$ is the field-dependent current density, which may include both linear (ohmic, for which $\mathbf{J} = \sigma \mathbf{E}$ where σ is the electrical conductivity) components and nonlinear (corona) components, $\mathbf{J}_C$ is the convection current density, which may include a contribution from precipitation, $\mathbf{J}_L$ is the lightning current density, and the last term is the displacement current density. In planar geometry, the current density $\mathbf{J}_M$ is the same at any height in the atmosphere, as required by the current continuity equation (e.g., Sadiku 1994). Krider and Musser (1982) suggested that the thundercloud, the postulated current source in the global electric

circuit, can be viewed as producing a quasi-static Maxwell current density even in the presence of lightning. In this view, the thunderstorm generator in the global electric circuit can, in principle, be monitored through its Maxwell current density.

The total Maxwell current density under thunderstorms has been measured by Krider and Blakeslee (1985), Deaver and Krider (1991), and Blakeslee and Krider (1992). Krider and Blakeslee (1985) found that the amplitude of $\mathbf{J}_M$ is of the order of 10 nA m^{-2} under active storms, while Deaver and Krider (1991) reported amplitudes of the order of 1 nA m^{-2} or less (in the absence of precipitation) under small Florida storms. Above thunderstorms, the convection and lightning terms, $\mathbf{J}_C$ and $\mathbf{J}_L$, in Eq. 1.2 are assumed to be negligible, and the Maxwell current density is expressed as the sum of only two terms:

$$\mathbf{J}_M = \mathbf{J}_E + \varepsilon_0(\partial \mathbf{E} / \partial t) \quad (1.3)$$

The Maxwell current density at altitudes of 16–20 km was measured by Blakeslee *et al.* (1989). They found that $\mathbf{J}_E$ typically accounted for more than half $\mathbf{J}_M$, while at the ground $\mathbf{J}_M$ is generally dominated by the displacement current density during active storm periods, as long as the fields are below the corona threshold of a few kV m^{-1} and there is no precipitation current. It is thought that, under some conditions, the Maxwell current density may be coupled directly to the meteorological structure of the storm and/or the storm dynamics (Krider and Roble 1986, pp. 5–6). However, the lack of simultaneous Maxwell current measurements both on the ground and aloft has prevented the details of this relationship from being determined.

1.4.5. Modeling of the global circuit

Models that can be used to calculate the electric field or the potential distribution around the thundercloud and the current that flows from a thundercloud into the global electric circuit have been developed by Holzer and Saxon (1952), Kasemir (1959), Illingworth (1972a), Dejnakintra and Park (1974), Hays and Roble (1979), Tzur and Roble (1985b), Nisbet (1983, 1985a, b), Browning *et al.* (1987), Hager *et al.* (1989a, b), Driscoll *et al.* (1992), Stansbery *et al.* (1993), Hager (1998), and Plotkin (1999). Both analytical and numerical models have been developed. Some models (e.g., Holzer and Saxon 1952; Hays and Roble 1979; Tzur and Roble 1985b) have been based on a quasi-static approximation, while others (e.g., Illingworth 1972a; Dejnakintra and Park 1974; Driscoll *et al.* 1992) have been designed to describe a time-varying problem that includes the effects of lightning. Nisbet (1983) modeled the atmosphere by a network of resistors, capacitors, and switches, where the current through these circuit elements represents conduction, displacement, and lightning currents, respectively. Sometimes corona cur-

rents have been taken into account (e.g., Tzur and Roble 1985b). Convection currents, including those due to precipitation, have usually been neglected. These models provide a convenient means of examining, through numerical experiments, the various processes operating in the global circuit. Normally, heights up to 100–150 km are considered, and thunderclouds are represented by two vertically displaced point charges of opposite polarity maintained by a current source. The upward-directed current from thunderclouds is assumed to spread out in the ionosphere of the storm hemisphere and, in some models (e.g., Hays and Roble 1979; Tzur and Roble 1985b; Browning *et al.* 1987; Stansbery *et al.* 1993) to flow along the Earth's magnetic field lines into the conjugate hemisphere. According to Stansbery *et al.* (1993), approximately half the current that reaches the ionosphere flows into the conjugate hemisphere and the other half flows to Earth in the fair-weather regions of the storm hemisphere. Pasko *et al.* (1998d) developed an electrostatic coupling model to examine the effects of the relatively steady electric fields of thunderclouds on the lower ionosphere.

1.4.6. Alternative views of the global circuit

Wilson's suggestion that thunderstorms are the main driving element in the atmospheric global circuit has been questioned by, among others, Dolezalek (1972), Williams and Heckman (1993), Kasemir (1994, 1996), and Kundt and Thuma (1999). Dolezalek (1988) suggested that the net charge on the Earth may be zero or near zero, rather than the $5 \times 10^5 \text{ C}$ postulated in the classical model of atmospheric electricity. Kasemir (1994, 1996) proposed an alternative concept of the global circuit in which the only equipotential layer is the Earth. He assumed that the very high negative potential of the Earth with respect to infinity drives the fair-weather current; lightning discharges and corona at the ground are claimed to be "local affairs" that do not contribute to the global circuit. In Kasemir's view, there are two types of generator in the global circuit, convection, acting in both stormy and fair-weather regions, and precipitation, both generators being driven by non-electrical forces. Kundt and Thuma (1999), who estimated a relatively low cloud top voltage, 1 MV relative to the Earth, asserted that thunderstorms are not important in the global electric circuit. However, Marshall and Stolzenburg (2001), from 13 balloon soundings of electric field through both convective regions and stratiform clouds, reported cloud top voltages ranging from -23 to $+79 \text{ MV}$, with an average of $+25 \text{ MV}$. The average cloud top voltage for the nine cases with positive values was $+41 \text{ MV}$. Marshall and Stolzenburg (2001) considered these voltage values as supporting Wilson's hypothesis that thunderstorms drive the global electric circuit. Further, they estimated an average cloud top voltage of $+32 \text{ MV}$ for

electrified stratiform clouds, this finding suggesting that stratiform clouds may make a substantial contribution to the global electric circuit.

1.5. Regarding the utilization of lightning energy

It appears to be impractical to utilize lightning energy. Each cloud-to-ground lightning flash involves an energy of roughly 10^9 to 10^{10} J (Section 4.2). For comparison, the energy required to operate five 100 W light bulbs continuously for one month is

$$5 \times 100 \text{ W} \times 3600 \text{ s} \times 24 \text{ h} \times 30 \text{ days} = 1.3 \times 10^9 \text{ J}$$

or about 360 kilowatt hours ($1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$), which is comparable to the total energy of one lightning flash. Even if it were possible to capture all the energy of a flash (this is not possible since the bulk of this energy is not delivered to the strike point but, rather, is lost to heating the air and producing thunder, light, and radio waves), one would need to attract 12 flashes to the energy storage facility in order to operate the five light bulbs for one year. The probability of a lightning strike to a given point on ground is very low. For example, a 1 m^2 area in Florida is struck by lightning on average, once in 10^5 years. A grounded structure protruding above Earth's surface is more likely to be struck by lightning. A 60 m tower located in Florida is expected to be struck by lightning at a rate roughly between once every other year and once every year (subsection 2.9.2). Thus, one needs 12 to 24 such towers covering a large area of 1 km^2 or so to operate the five 100 W light bulbs, which is obviously impractical. Most of the United States experiences lightning activity that is a factor 2 to 3 lower than in Florida. As a result, the number of lightning-capturing towers needed to operate the five 100 W bulbs in areas of moderate lightning activity would be 24 to 72.

Thus the three main problems with the utilization of lightning energy (leaving aside the issue of energy storage devices) can be formulated as follows.

- (i) The power associated with a lightning flash is very high, but it is released in pulses of very short duration (of the order of 10^{-4} – 10^{-5} s). As a result, the lightning energy, the integral of the power over the short period of time, is moderate, comparable to the monthly energy consumption, 360 kilowatt hours, of five 100 W light bulbs.
- (ii) Not all the lightning energy in a flash is delivered to the strike point. Using a typical value of energy per unit resistance (action integral) of 10^5 A s^2 (Table 4.4) determined from measurements of the current at the negative lightning channel base and an assumed range of resistances at the strike point of 10 to 100Ω , we estimate the range of the lightning

energy delivered to the strike point to be from 10^6 to 10^7 J , which is only 10^{-2} to 10^{-4} of the total energy.

- (iii) The capturing of a sufficiently large number of lightning strikes would require the use of a large number of tall towers, which is impractical.

1.6. Summary

In ancient cultures lightning was viewed both as a weapon of the gods, used by them to punish humans, and as a message from heaven used to influence state affairs. Systematic studies of thunderstorm electricity began in 1752 when an experiment proposed by B. Franklin was conducted in France. The results of that experiment and similar following experiments in other countries showed conclusively that thunderclouds were electrified. In the same year that the first thunderstorm electricity experiment was carried out, fair-weather electricity was discovered. According to the “classical” view of atmospheric electricity, the potential of the electrosphere, located at an altitude of 60 km or so, with respect to the Earth's surface is about 300 kV. This potential difference drives a total conduction current of the order of 1 kA in all fair-weather regions on the globe. Thus, the global fair-weather load resistance is about 300Ω . Thunderstorms are considered as the sources in the global electric circuit. Since there has been no experiment to confirm conclusively this classical picture of atmospheric electricity, it remains a subject of debate. The salient properties of downward negative lightning discharges, the most common type of lightning, are summarized in Table 1.1. It appears to be impractical to utilize lightning energy.

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2.1. Introduction

More than 75 years ago Brooks (1925) estimated that the global lightning flash rate, the total of both cloud and ground flashes, was about 100 s^{-1} . He did so by combining the following information: (i) an estimated average of 16 storms per year over the assumed observation area, 512 km^2 , of an individual weather station, which was extrapolated to the entire Earth's surface, about $5.1 \times 10^8 \text{ km}^2$ (both land and sea included); (ii) an assumed average storm duration of 1 hr; and (iii) the flash rate of 3.5 min^{-1} observed by Marriott (1908) during a 28 min thunderstorm period in England. More recent estimates of the global flash rate, based on satellite measurements and many assumptions, are within a factor 2 or 3 of this value but are not necessarily more accurate (subsection 2.5.3). The number of flashes per unit area per unit time is called the flash density, and the number of ground flashes per unit area per unit time (usually per square kilometer per year: Section 2.5) is called the ground flash density. A global flash rate of 100 s^{-1} corresponds to a total flash density over the entire Earth's surface of about $6 \text{ km}^{-2} \text{ yr}^{-1}$ and a ground (land or sea) flash density of about $1.5 \text{ km}^{-2} \text{ yr}^{-1}$ if we assume a 3 : 1 global ratio of cloud to cloud-to-ground flashes (Section 2.7). Lightning activity is highly variable on both (i) a temporal scale at a given location, ranging from some tens of years (Section 2.6) to the lifetime of individual thunderstorm cells, which is of the order of one hour (Chapter 3) or shorter, and (ii) a spatial scale ranging from continents and oceans to a few kilometers or less. The ground flash density N_g is the basic information needed for estimation of the lightning strike incidence to a ground-based object or system. Other measures of the lightning incidence to an area include the annual number of thunderstorm days T_D (Section 2.3) and the annual number of thunderstorm hours T_H (Section 2.4). Various characteristics of cloud-to-ground lightning such as the number of strokes per flash

The worldwide distribution of lightning remains an unknown despite the advancement of ground-based and space-based lightning detection systems . . . Only part of the land is now covered by lightning detection networks; and the first satellites to measure lightning will be in low orbit, thus measuring only part of the global lightning.

R.E. Orville (1995)

and the polarity of the charge lowered to ground apparently depend on season, location, and storm type (Section 2.8). Occurrence statistics and various characteristics of positive lightning are considered further in Chapters 5 and 8.

Clearly, T_D , T_H , and N_g each vary both in space (from one location to another) and in time (e.g., from year to year). Accordingly, there are two different forms in which these quantities are presented, which emphasize either the spatial or the temporal component of their variability. The spatial variability is usually presented in the form of a map for many locations, the temporal variability being suppressed typically by the use of long-term average values (see, for example, Figs. 2.8, 2.9, and 2.11). The temporal variability is usually presented in the form of a time series for a single location (see, for example, Figs. 2.13 and 2.14). When considering correlations among T_D , T_H , and N_g , one should distinguish between these two components of variability.

2.2. Characterization of individual storms and storm systems

2.2.1. Lightning flash rate

Cianos and Pierce (1972) stated that the average flash rate (also called the flashing rate) during the lifetime of any thunderstorm cell (Section 3.1) is about 3 min^{-1} , irrespective of locality. This statement is apparently supported by the finding of Boccippio *et al.* (2000a) that ocean storms produce lightning at rates that are not dissimilar from those of continental storms. Livingston and Krider (1978) gave statistical data on lightning at the Kennedy Space Center (KSC), Florida for 22 storm days during 1974 and 1975. The data were derived from the electric fields recorded by the KSC field mill network (subsection 17.2.1). The number of cloud and ground discharges per overall storm (individual storm cells could not, in general, be identified) ranged from eight to 1987. About 70 percent of all lightning occurred during what was termed the “active

storm period”, which represented only about 30 percent of the total storm duration. A strong correlation was found between the logarithm of the total number n of discharges during the active period and the duration in minutes D_A , of that period. The regression equation fitting the 23 data points is

$$\log_{10} n = 0.01 D_A + 1.90 \quad (2.1)$$

The average flash rate per overall storm varied between 0.3 and 9.3 min^{-1} . On five of the storm days, the flash rate averaged over 5 min intervals exceeded 20 min^{-1} , the maximum rate being 26 min^{-1} . Small storms, which perhaps can be regarded as single-cell storms, produced a maximum flash rate of typically 1 to 4 min^{-1} . Piepgrass *et al.* (1982), also using the KSC field-mill network, presented data on the frequency of lightning occurrence for 79 additional KSC storms during the summers of 1976 through 1980. The number of discharges per storm varied from 1 to 3687. Average flash rates for the storms were between 0.1 and 12.2 min^{-1} . Maximum flash rates, averaged over 5 min intervals, were between 0.2 and 30.6 min^{-1} .

Lhermitte and Krehbiel (1979) reported an abrupt increase in the flash rate from about 10 to 60 min^{-1} in a “normal-size” KSC storm. The discharges were identified using the VHF time-of-arrival source location system called LDAR (subsection 17.4.2). The primary contributors to the increase in flash rate were intracloud discharges that originated at heights ranging from 8 to 10 km (-20 to -35°C) above the area of maximum radar reflectivity ($> 50 \text{ dBZ}$) and were apparently due to the development of a strong updraft and the associated precipitation growth.

Maier and Krider (1982) used a two-station lightning locating system based on the wideband magnetic field direction finding (DF) technique (Section 17.3) to study the occurrence of cloud-to-ground lightning in three severe storms in northern Texas and Oklahoma and 268 air-mass storms (Section 3.1) in south Florida. Severe storms are defined by the US Weather Service as storms that produce at least one of the following three conditions: (i) large hail (diameter $\geq 1.9 \text{ cm}$), (ii) damaging winds (speed $\geq 26 \text{ ms}^{-1}$), or (iii) tornadoes (e.g., Johns and Doswell 1992). Note that lightning is not part of this definition. Sometimes storms are subjectively labeled as severe without reference to the formal definition given above. The three severe storms analyzed by Maier and Krider (1982) produced tornadoes, and therefore they satisfied the U.S. Weather Service criteria. The severe thunderstorms in Maier and Krider’s analysis lasted from 2 to 8 hr and produced 200 to 2800 cloud-to-ground flashes over areas of 2000 to 5000 km^2 . The Florida air-mass storms rarely lasted longer than 1 to 2 hr, producing about 50 discharges over an area of about 450 km^2 . However, there were usually only one or two severe storms per storm day in the Texas–Oklahoma region while in Florida there were 10 to 50 storms per storm day. The severe

Table 2.1. Means and standard deviations (the latter in parentheses) of various storm parameters studied by Peckham *et al.* (1984)

Parameter	Single-peak storms	Multiple-peak storms	Storm systems
Duration (min)	41 (16)	77 (26)	130 (51)
Area (km^2)	103 (63)	256 (154)	900 (841)
Number of ground flashes	73 (69)	270 (216)	887 (720)
Ground flash density ($\text{km}^{-2} \text{ min}^{-1}$)	0.018 (0.011)	0.015 (0.008)	0.010 (0.006)
Average ground flash rate (min^{-1})	1.7 (1.2)	3.4 (2.3)	6.8 (4.7)
Maximum ground flash rate (min^{-1})	3.7 (2.6)	7.3 (4.4)	14 (9)

storms produced cloud-to-ground lightning at a mean rate of about 2 to 6 min^{-1} , while the air-mass storm rate was about 1 min^{-1} . The peak ground flash rate in severe storms approached 20 min^{-1} , averaged over 5 min intervals, while in the 268 air-mass storms it was about 12 min^{-1} . Apparently no correction was applied to the data to account for the lightning locating system’s estimated detection efficiency, about 80 percent (Section 17.3).

Peckham *et al.* (1984) used the same type of ground flash locating system as Maier and Krider (1982) to study cloud-to-ground lightning in the Tampa Bay area of Florida. For 111 storms on eight days, the following parameters were determined: duration, area, number of ground flashes, average ground flash density, average ground flash rate, and maximum ground flash rate averaged over a 5 min period. Note that the flash density definition used in this section to characterize an individual storm or parts of a storm system is different from that used below in Section 2.5 to characterize lightning incidence to an area. The means and standard deviations of the six quantities analyzed by Peckham *et al.* (1984) are presented in Table 2.1 where the storms are grouped into three categories: (i) single-peak storms, i.e., those with a single peak in their flash rate versus time curves, (ii) multiple-peak storms, i.e., those with multiple-peak flash rates, and (iii) storm systems composed of groups of the first two types of storm, which would be detected as one overall storm by a nondirectional lightning detector such as a lightning flash counter (subsection 2.5.1) or perhaps a network of electric field mills (subsection 17.2.1). Peckham *et al.* (1984) defined the storm duration as the period of the ground flash rate curve during which there is no more than one 5 min interval without lightning. A map of one day’s ground flash activity is shown in Fig. 2.1. The

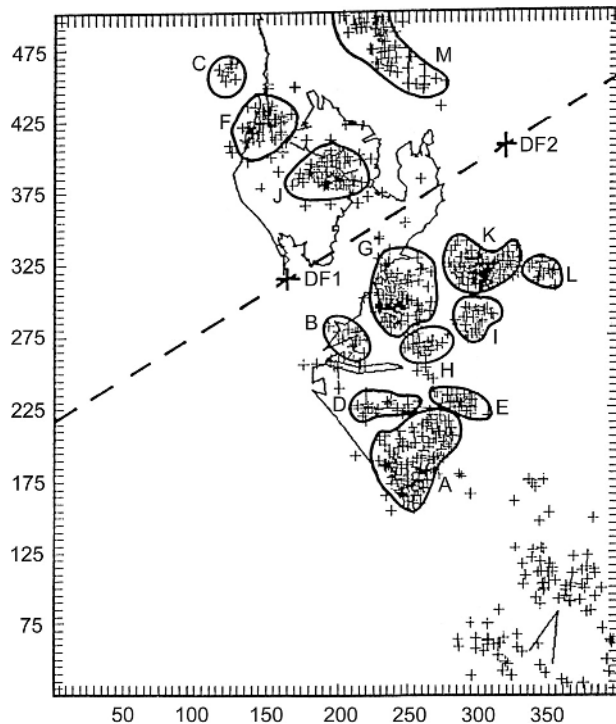


Fig. 2.1. Map of ground flash locations in the Tampa Bay area (Florida) on 8 August 1979 between 13:00 and 16:00 EDT. The two direction finders comprising the lightning locating system are labeled DF1 and DF2. The scale units are thousands of feet (1000 ft = 305 m). Adapted from Peckham *et al.* (1984).

letters indicate the chronology of the storm starting time, including the single- and the multiple-peak storms. Two storm systems are composed of storms ABDEGHIKL and storms CFJ. As can be seen in Table 2.1, Peckham *et al.* (1984) found an average ground flash density for storm areas containing lightning of about $0.01 \text{ km}^{-2} \text{ min}^{-1}$ (an underestimate due to the less than 100 percent detection efficiency). They found that the number of ground flashes, n_g , in a storm and the storm duration, D_s , in minutes were related by

$$\log_{10} n_g = 0.014 D_s + 1.2 \quad (2.2)$$

The corresponding scatter plot is shown in Fig. 2.2.

The maximum flash rates in different storms and storm systems, as reported by various investigators, were summarized by Goodman and MacGorman (1986) and are given in Table 2.2. Most of the maximum flash rate values, some of which are for both cloud and ground lightning discharges while others are for ground discharges only, are in the range from a few to some tens per minute. Recent VHF observations indicate the existence of previously unknown forms of cloud discharge activity in storms in the US Great Plains. Specifically, Krehbiel *et al.* (2000) observed, in a large supercell storm, what they describe as essentially

continuous discharge activity during the 1 min time interval presented. The activity filled a saucer-shaped volume with a diameter over 60 km and with a vertical extent from about 4 to 12 km.

2.2.2. Lightning in the Blizzard of '93

Orville (1993), using data from the US National Lightning Detection Network (NLDN) (Section 17.5), published the characteristics of cloud-to-ground lightning activity in a storm that was termed the Blizzard of '93. The storm occurred on 12–14 March 1993, developed over Texas and the Gulf of Mexico and then moved over Florida and up the US East Coast, breaking records for snowfall and low pressure as it moved toward the Canadian Maritime region. In the 48 hr period from 06 UT 12 March to 06 UT 14 March 1993, this cold-season storm produced over 59 000 cloud-to-ground flashes with a peak lightning rate exceeding 5100 hr^{-1} . For this storm, ground flash densities were in excess of 0.16 km^{-2} in an area just south of Tampa, Florida, while few flashes to ground were recorded north of the Carolinas. The overall percentage of positive flashes averaged about 13 percent, but positive flash percentages exceeded 60 percent in the northern portion of the storm. Estimated median peak currents for negative flashes were 30 kA and for positive flashes 52 kA. A summary of the cloud-to-ground lightning activity for the Blizzard of '93 is plotted in Fig. 2.3 for the period from 16:02 UT 12 March to 10:02 UT 14 March 1993 (52 hours).

2.2.3. Lightning in hurricanes

Molinari *et al.* (1994), using NLDN data and infrared satellite images, studied the spatial and temporal distribution of cloud-to-ground lightning in the 1992 Hurricane Andrew, one of the most intense hurricanes in United States history. A distinct radial variation was observed in the time-averaged flash density, with a weak maximum in the eye wall, a region of near zero flash density 40 to 100 km from the center, and a steady increase to a large maximum in the outer rainbands (regions of heavier precipitation) 190 km from the center. Similar findings were reported by Lyons and Keen (1994), who additionally noted near-eyewall bursts of cloud-to-ground flashes (detected with a TOA lightning locating system: subsection 17.4.3) preceding periods of intensification in Hurricanes Diana (1984) and Florence (1988). Samsury and Orville (1994), who studied the 1989 Hurricanes Hugo and Jerry, each over an 18 hr period, found that over half the 33 cloud-to-ground flashes in Hurricane Hugo occurred in or near the eyewall, while the outer convective rainbands were the preferred locations of the 691 flashes during Hurricane Jerry. Williams (1995) presented lightning locations for Hurricane Andrew that coincided with

Table 2.2. Maximum flash rates in various types of storms. Adapted from Goodman and MacGorman (1986)

Reference	Location	Storm type	Maximum rate, min ⁻¹
Shackford (1960)	New England	hailstorms	30 (1800–60 min)
Mackerras (1963)	Subtropics	—	120
Blevins and Maurwitz (1968)	Great Plains	hailstorms	70
Fuquay and Baughman (1969)	Northwest United States	—	23
Israel (1973)	—	—	67 (1000/15 min)
Kinzer (1974)	Oklahoma	squall line	25 (5594/225 min) ^a
Livingston and Krider (1978)	Florida	tornado	26
Piepgrass <i>et al.</i> (1982)	Florida	isolated storms or clusters	7.6 (5 min average) ^b
Peckham <i>et al.</i> (1984)	Florida	isolated storms or clusters	3.7–14 (5 min average) ^a
Johnson and Goodman (1984)	Gulf of Mexico	Hurricane Alicia	53 (3150/60 min)
Goodman and MacGorman (1986)	Oklahoma	MCC	45 (1 h average) ^a
			54 (1 h maximum) ^a
			60 (5 min maximum) ^a

^a Ground discharges only.
^b Range from 0.6–30 min⁻¹.

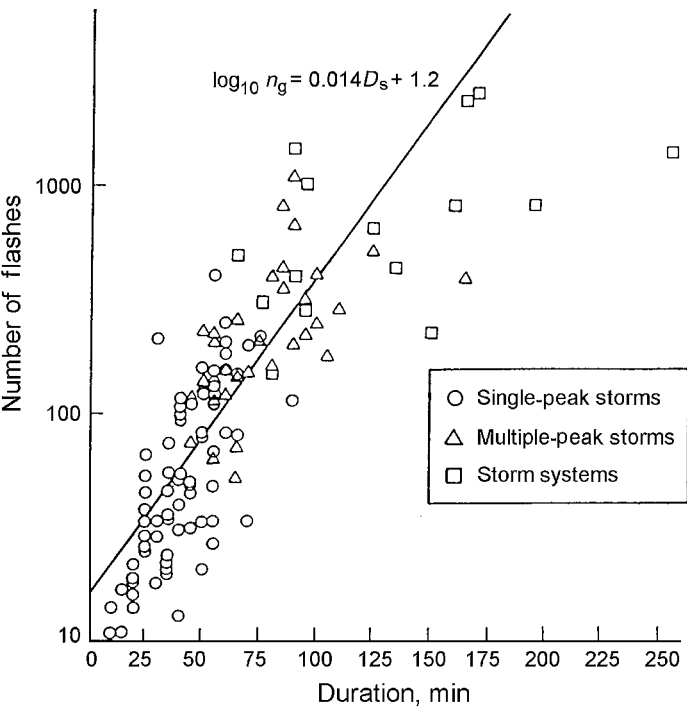


Fig. 2.2. Number of ground flashes versus storm duration in the Tampa Bay area, Florida. In determining the regression line (correlation coefficient $r = 0.78$), the storm system data were excluded; nevertheless these are plotted because the storm systems are a combination of single- and multiple-peak storms and because their definition is subjective. Interestingly, the storm system data lie near the regression line. Adapted from Peckham *et al.* (1984).

eyewall convection as this storm crossed the warm waters of the Gulf Stream toward the Florida coast. The most extensive study to date of cloud-to-ground lightning in hurricanes was conducted by Molinari *et al.* (1999). A total of nine Atlantic basin hurricanes were examined, four of

which had been studied previously by Samsury and Orville (1994) or Lyons and Keen (1994). A common radial distribution of ground flash density, generally similar to that described earlier in this section, was found: a weak maximum in the eyewall region, a clear minimum 80–100 km

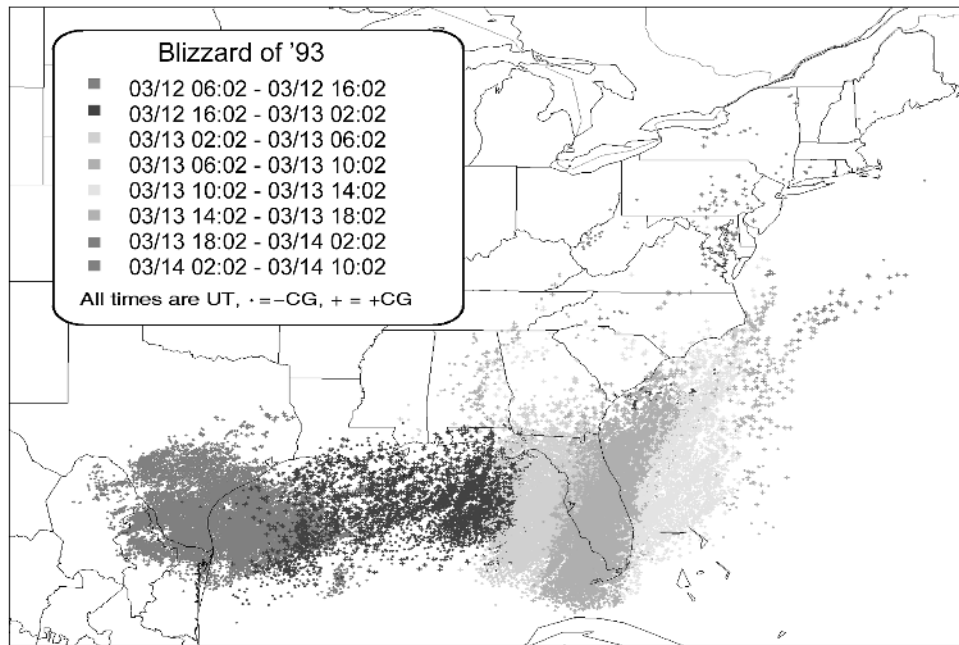


Fig. 2.3. See also the colour plate section. A summary of the cloud-to-ground lightning activity reported by the NLDN for the Blizzard of '93, from 06:02 UT 12 March to 10:02 UT 14 March 1993. The total number of recorded flashes for this time period was 62 887. The dots indicate negative flashes (55 206) and the plusses indicate positive flashes (7681). Different colors correspond to different times, as shown. Courtesy of Global Atmospheric, Inc.

outside the eyewall, and a strong maximum in the vicinity of the outer rainbands (at a distance of 210–90 km from the center). Molinari *et al.* (1999) considered this flash density distribution as supportive of the division of the convective structure of hurricanes into three distinct regions: the eyewall, the inner bands, and the outer bands. The eyewall is a unique phenomenon but shares some attributes with deep, weakly electrified oceanic monsoonal convection (see, for example, Williams 1995). The inner bands have characteristics similar to the trailing stratiform region of mesoscale convective systems (see subsection 2.2.4), including the production of a relatively high fraction of positive flashes. The outer bands contain the vast majority of the ground flashes in hurricanes.

2.2.4. Lightning in mesoscale convective complexes

Goodman and MacGorman (1986), using the four-station National Severe Storms Laboratory (NSSL) DF lightning locating network (Section 17.3), analyzed lightning discharges to ground in 10 mesoscale convective complexes (MCCs). The MCC is a type of mesoscale convective system (MCS), the latter being described by Goodman and MacGorman (1986) as a weather system having a horizontal extent of 250–2500 km, occurring on a time scale in excess of a few hours, and including significant convection during some part of its life cycle. Examples of MCSs are squall lines, tropical cyclones, and hurricanes. MCSs

are discussed in detail in MacGorman and Rust (1998, Section 8.2) and MacGorman and Morgenstern (1998). They are also discussed in relation to the lightning “bipolar pattern” in subsection 2.2.7 below. During the most intense phase of its life cycle, a single MCC can produce one-fourth of the mean annual lightning strikes to ground at any site it passes over. For that reason MCCs, although a relatively infrequent meteorological situation, are considered here in a separate subsection in our characterization of various types of storms and storm systems.

MCCs were identified by Goodman and MacGorman (1986) on the basis of their circular appearance, long duration, and large size of the cloud shield in Geostationary Operational Environmental Satellite (GOES) infrared images. Goodman and MacGorman (1986) found that a typical MCC is characterized by a cloud-to-ground flash rate in excess of 1000 hr⁻¹ for nine consecutive hours. Peak ground discharge rates of 60 min⁻¹ were not uncommon, the average being 42 min⁻¹. They also compared their results with maximum flash rates published by others (Table 2.2).

2.2.5. Lightning and severe weather phenomena

As defined in subsection 2.2.1, severe weather phenomena include large hail (diameter ≥ 1.9 cm), damaging winds (speed ≥ 26 m s⁻¹), and tornadoes. Reap and MacGorman (1989), using 1985–86 data from the NSSL

network (Section 17.3) for the Great Plains of the United States, found a significant correlation between the occurrence of severe weather and elevated rates of 30 or more positive ground flashes per hour within $48 \text{ km} \times 48 \text{ km}$ grid blocks. Also, Rust *et al.* (1985a), Curran and Rust (1992), Branick and Doswell (1992), Seimon (1993), MacGorman and Burgess (1994), Stolzenburg (1994), and Carey and Rutledge (1998) noted that severe weather occurred in storms in which ground flashes had predominantly positive polarity during significant portions of the storm. Near the beginning of active cloud-to-ground lightning periods of hail-producing thunderstorms in the Great Plains of the United States, Stolzenburg (1994) reported positive lightning flash rates as high as 67 flashes in 5 min and flash densities up to $0.6 \text{ km}^{-2} \text{ hr}^{-1}$. Nevertheless, many severe storms do not exhibit large numbers of positive ground flashes, and positive flashes are known to occur in non-severe storms (Chapters 5 and 8). Further, there is considerable variability in the evolution of the positive ground flash rate relative to storm evolution and severe weather occurrence (MacGorman and Burgess 1994; Stolzenburg 1994).

Williams *et al.* (1999) studied lightning activity in severe thunderstorms in Florida. Thunderstorms were identified as severe on the basis of reports by surface observers of dime-size or greater hail, strong winds (trees blown down), or the occurrence of tornadoes. The associated ground flash rates were obtained using NLDN data, and total flash rates were estimated from light detection and ranging (LDAR) records (subsection 17.4.2). In the latter case, any detected VHF source that occurred within 300 ms and 5 km of a previous source was assigned to the same flash as the previous source, the maximum allowed flash duration being 5 s. For more than 10 percent of so-defined flashes only one VHF source was detected. Clearly, in such a grouping of VHF sources into flashes, the decomposition of single large flashes into several smaller ones is possible, the result being an overestimation of the total flash rate. Williams *et al.* (1999) found that severe storms in Florida were characterized by relatively high maximum total flash rates ranging from about 60 to 500 min^{-1} . The most likely maximum total flash rates associated with small, non-severe thunderstorms in the same study were reported by Williams *et al.* (1999) to be in the range of 1 to 10 min^{-1} . The maximum ground flash rate was typically more than an order of magnitude lower than the maximum total flash rate. The polarity of ground flashes was not specified by Williams *et al.* (1999). Various aspects of severe storms, including the production of lightning, are reviewed in Williams (2001).

Changnon (1992) examined ground flash rates and strike locations relative to 48 streaks of crop damage produced by hail (regardless of its size; possibly smaller

than required for a storm to be formally classified as severe) in Illinois during six days in June–July 1989. All ground flashes associated with hail streaks were of negative polarity. Ground flashes from hail-producing cells typically began 8–10 min before the first hail damage and exhibited a flash rate and area of coverage that grew rapidly until damage began. Then the ground flash rate decreased, but remained large during hail damage, and decreased more rapidly after hail damage ceased. Ground flash activity typically ended 8 min after the last hail damage. Shafer (1990) observed a storm system in Oklahoma that produced large hail before producing ground flashes.

MacGorman and Rust (1998) summarized the information on lightning activity in tornadic storms. Both total and cloud flash rates in tornadic supercell storms (a supercell can be defined as a long-lived cell with a rotating updraft) were observed to be large at the time of tornado occurrence (MacGorman *et al.* 1985, 1989; Richard 1992). However, the observed relationship between a tornado and the ground flash rate is rather variable. In three tornadic storms studied by Orville *et al.* (1982), MacGorman *et al.* (1989), and Richard (1992), ground flash rates were small before and during tornadoes. In four, studied by Kane (1991), MacGorman and Nielsen (1991), Keighton *et al.* (1991), and Seimon (1993), ground flash rates peaked during tornadoes, but were usually near a local minimum when the tornadoes began. In two storms studied by MacGorman *et al.* (1985) and Kane (1991), tornadoes occurred after ground flash rates had begun to decrease from their peaks but before they had decreased to a local minimum. Thirty-four of the 42 tornadoes produced on four days and studied by MacGorman and Burgess (1994) began when ground flash rates were either $< 0.5 \text{ min}^{-1}$ or near a local minimum value. During seven of the 10 tornadoes that lasted $\geq 20 \text{ min}$, ground flash rates increased while the tornado was occurring. Similar variability was observed by Knapp (1994), who analyzed ground flashes in 264 storms that produced tornadoes east of the Continental Divide of the United States on 35 days from 18 April through 1 July 1991. Perez *et al.* (1997), using NLDN data, studied cloud-to-ground lightning patterns in 42 tornado-producing supercells that occurred between January 1989 and November 1992. Thirty-one (74 percent) of the 42 storms were characterized by a peak in the ground flash rate preceding tornado formation. Twenty storms (48 percent) exhibited a local minimum ground flash rate coincident with tornado formation. Six of the 42 storms showed a reversal, from positive to negative, in the sign of the charge lowered to ground. Smith *et al.* (2000) inferred, from studying three tornadic storm events, that the lightning polarity reversals were influenced by the changes in intensity of the storms' updrafts brought about by changes in the buoyancy of ingested surface and/or boundary layer air.

Lightning activity in a tornadic storm in central Oklahoma, as revealed by a VHF TOA lightning imaging system (subsection 17.4.2), is presented by Krehbiel *et al.* (2000). The lightning activity exhibited a counterclockwise “hook” structure in the vicinity of the tornado, similar to the hook echoes observed by meteorological radars in tornadic storms. Lightning activity within about 3 min of a tornadic thunderstorm observed from space by the optical transient detector (OTD) (Section 17.8) was reported by Buechler *et al.* (2000).

Most modern studies of lightning in relation to damaging wind in isolated thunderstorms have been concerned with microbursts. These transient strong downdrafts (having a life cycle of 15 to 20 min) produce horizontal winds near the ground that are directed outward from the downdraft and may form one or more vortex rings around the downdraft. The horizontal outflow region is typically 2–4 km across, and the vortex rings may rise about 700 m above the ground. The microburst is particularly hazardous to aircraft during take off or landing. It is worth noting that in most cases the speeds of microburst winds are lower than required for a thunderstorm to be formally classified as severe (see above; Williams *et al.* 1999).

MacGorman and Rust (1998) summarized the available information on the lightning–microburst relation as follows. In the southeastern United States microbursts are often accompanied by moderate to heavy rain, while in the drier environment of the western United States sometimes they are accompanied only by light rainfall. Goodman *et al.* (1988a, b) and Williams *et al.* (1989c) found that microbursts in the southeastern United States tended to occur roughly 10 min after the peak in the total lightning flash rate and shortly after, or roughly coincident, with the peak in the ground flash rate. The storm studied by Goodman *et al.* (1988b) produced a total of 116 flashes, only six of which were discharges to ground. The peak flash rate, 23 min^{-1} , occurred 7 to 8 min after the initial discharge and 4 min prior to the microburst onset, simultaneously with the peaks in the storm mass, vertically integrated liquid water content, echo volume, and cloud height. The maximum rain flux and rain rate occurred 2 min after the peak flash rate. For the case of microbursts associated with moderate to heavy precipitation, similar lightning–microburst relationships were found in the western United States (Williams *et al.* 1989b) and in the northeastern United States (Kane 1991). However, Williams *et al.* (1989b) also observed that storms producing microbursts associated with light precipitation exhibited much smaller lightning flash rates than did the storms with heavier precipitation. While lightning preceded the light-precipitation microbursts, there was no pronounced peak in total flash rate prior to these “dry” microbursts, many of which occurred without any lightning in the parent storms.

2.2.6. Flash rate versus some non-electrical cloud properties

Many investigators have related flash rates to cloud properties determined from radar observations. We review the most important results of those studies; further information can be found in MacGorman and Rust (1998). Some of the results appear to contradict each other, but no attempt is made here to resolve these controversies.

Jacobson and Krider (1976) and Livingston and Krider (1978) showed that total flash rates greater than 10 min^{-1} in Florida are associated only with storms having maximum radar echo tops above 14 km. At the Kennedy Space Center, a total flash rate of 10 min^{-1} is probably indicative of a ground flash rate of roughly 4 min^{-1} (Livingston and Krider 1978). Holle and Maier (1982) reviewed the worldwide literature regarding the cloud top temperature at or after the time of the first lightning. They also reported that the minimum radar echo height (generally assumed to represent the thundercloud height) at which ground flashes were observed in Florida was 7.8 km. Larsen and Stansbury (1974), Marshall and Radhakant (1978), and Cherna and Stansbury (1986) showed that the total flash rate (represented by the occurrence rate of atmospherics at a single station) in Canada was correlated with the area of the storm as measured by radar. The dominant parameter determining the rate of electrical activity, according to Cherna and Stansbury (1986), is the thundercloud height. Goodman *et al.* (1988b) reported that the peak flash rate in a microburst-producing storm occurred simultaneously with the radar-determined peak in storm mass, vertically integrated liquid water content, echo volume, and cloud height. Solomon and Baker (1994) reported that lightning was correlated with the maximum radar-determined cloud height in New Mexico storms, no lightning being observed in clouds whose top heights did not exceed approximately 9.5 km. Lhermitte and Krehbiel (1979), using a network of three Doppler radars and the LDAR system (subsection 17.4.2), demonstrated that the total lightning flash rate was correlated with in-cloud updraft velocity. Williams (1985) gave a relationship between total flash rate F and cloud height H based on observations in Florida, New Mexico, and New England (see Fig. 2.4), $F = 3.44 \times 10^{-5} H^{4.9}$. Price and Rind (1992) used this relationship for continental thunderstorms and a different one for maritime storms to infer global lightning distributions from satellite-observed cloud heights. Molinié and Pontikis (1995), however, found no correlation between the total flash rate estimated using the SAFIR interferometer (Section 17.6) and the cloud top height inferred from daily meteorological sounding in the coastal region of French Guyana. Michalon *et al.* (1999) expressed the flash rate as a power function of both the cloud top height and the cloud droplet concentration.

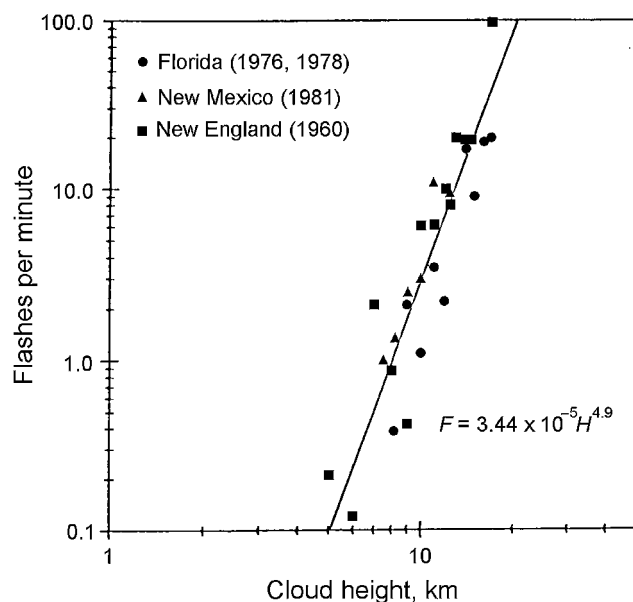


Fig. 2.4. Relationship between lightning flash rate F and convective cloud top height H . Adapted from Williams (1985). Note that a fifth-power relation between F and H had been predicted previously by Vonnegut (1963).

Reap (1986) compared cloud-to-ground flash rates from the BLM network (Section 17.3) with GOES-determined cloud top temperatures and reported a tendency for cloud-to-ground flashes to be clustered under or near the coldest, that is, the highest, cloud tops. Watson *et al.* (1995), using data from both DF (Section 17.3) and TOA (subsection 17.4.3) lightning locating systems, studied the relationships between cloud-to-ground lightning and (i) the heights of radar echo tops, (ii) the radar-determined vertically integrated liquid water content. The greater percentage of echoes with lightning occurred when the echo-top heights exceeded about 15 km, and this percentage decreased with decreasing echo-top heights. Holle and Watson (1996), using the NLDN, studied winter storms over the central United States and found that often regions with cloud-to-ground lightning could be better identified by high echo tops than by reflectivity (the radar reflectivity of particles found in clouds is discussed, for example, in subsection 7.1.2 of MacGorman and Rust 1998). However, Petersen *et al.* (1996), who studied convection and associated cloud-to-ground lightning activity over the tropical western Pacific Ocean, reported that tropical oceanic clouds whose tops were often above 15 km produced little or no lightning (see also Williams *et al.* 1992; Rutledge *et al.* 1992; and subsection 2.2.8 below). They attributed this feature to the fact that in such convection the most significant hydrometeor mass, as evidenced by reflectivities greater than 30 dBZ (the reflectivity Z expressed in dBZ is $10 \log_{10} Z$), was often confined to elevations

below the height of the freezing level, a condition not conducive to electrification processes (subsection 3.2.6). The oceanic convection that did produce significant lightning activity had reflectivities in excess of 30 dBZ in the mixed-phase region, and the cloud-to-ground flash rate was observed to increase simultaneously with the descent of the precipitation mass bounded by the 30 dBZ reflectivity contour.

Using a polarimetric radar (see, for example, subsection 7.1.2 of MacGorman and Rust 1998), Carey and Rutledge (1996) found that cloud flash rates were proportional to the graupel/hail volume (graupel is millimeter-size soft hail) at temperatures below 0 °C, while for one Colorado storm ground flash rates were proportional to the graupel/hail volume at temperatures higher than 0 °C. Lopez and Aubagnac (1997) found, for an Oklahoma hail-storm with well-developed supercell organization, that the development of graupel above the freezing level was related to the overall increase and decrease in the production of ground flashes, while the descent of small hail (less than 2 cm in diameter) below the height of the graupel region was responsible for the three or four brief increases in ground lightning activity. Carey and Rutledge (2000) reported that, in tropical island convection, the surface electric field, the total lightning flash rate, and the cloud-to-ground lightning flash rate were highly correlated with the radar-inferred ice mass in the mixed-phase region.

MacGorman and Rust (1998: p. 225) noted that a lightning-producing cloud is a very complex system whose electrical state is difficult to determine from measurements of its nonelectrical properties.

2.2.7. Lightning bipolar pattern

Orville *et al.* (1988) were the first to report on thunderstorm systems that produce positive and negative ground flashes whose ground strike locations tend to be separated in space, forming a so-called bipolar pattern. An example is given in Fig. 2.5. Orville *et al.* (1988) gave the following characteristics of the bipolar pattern, which occurs in many MCSs (defined in subsection 2.2.4). The pattern has a characteristic length of about 100 km. It is aligned with the upper-level winds and has been observed in all seasons but seems to occur more frequently in the fall and winter. The ratio of the positive flash density (the number of positive ground flashes per unit area accumulated over the life of the storm) to the negative flash density is approximately 0.1 with a range from 0.05 to 0.2. Orville *et al.* (1988) described the bipolar pattern as a persistent feature that, for the case characterized by them in detail, existed for 11 hours and was oriented from northeast (positive pole) to southwest (negative pole) as the storm moved to the east. The various studies reviewed below have revealed that the positive flashes in the bipole occur predominantly in the latter stage

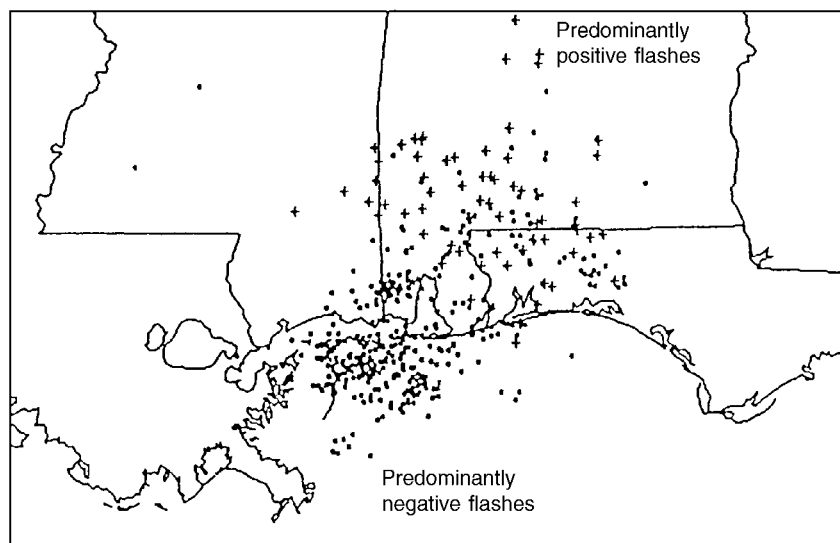


Fig. 2.5. An example of lightning bipolar pattern formed during one hour on 22 February 1987 along the Gulf of Mexico coast of the United States. Positive flashes are shown by plus signs and negative flashes by squares. Adapted from Orville *et al.* (1988).

of the lifetime of the storm system, so that the change in the predominant flash polarity combined with storm motion can produce the observed spatial bipole. Further, positive flashes are generally associated with regions of relatively shallow convection, such as the trailing stratiform region behind the convective line in summer MCSs and the shallower clouds downwind of the large-scale geostrophic flow (e.g., Houghton 1997) in both winter and summer MCSs. Apparently a bipolar pattern can be also formed by two (or more) isolated storms several tens of kilometers apart, each dominated by a different polarity of ground flashes (e.g., Engholm *et al.* 1990). Thus it appears that the lightning bipolar pattern is not specific to an MCS or any other type of storm but can occur in a variety of situations.

Examples of winter bipolar patterns over both land and ocean are given by Brook *et al.* (1989). The over-land storm moved northeast for a period of about 14 hours, the positive strokes occurring in the latter part of the storm life cycle. The resultant spatial bipole, produced via (i) the change of predominant flash polarity during the storm life cycle and (ii) the relatively fast motion of the storm, is oriented in such a way that the positive pole is northeast of the negative pole.

Rutledge and MacGorman (1988) used the NSSL DF lightning locating network (Section 17.3) in conjunction with both conventional and Doppler radar data over a 6 hr period to study the 10–11 June 1985 Kansas MCS. The storm consisted of a 20-km-wide line of convective cells (known as a squall line) trailed by an 80-km-wide stratiform precipitation region and exhibited the bipolar pattern described above, the negative lightning activity being associated mostly with the convective region. The negative

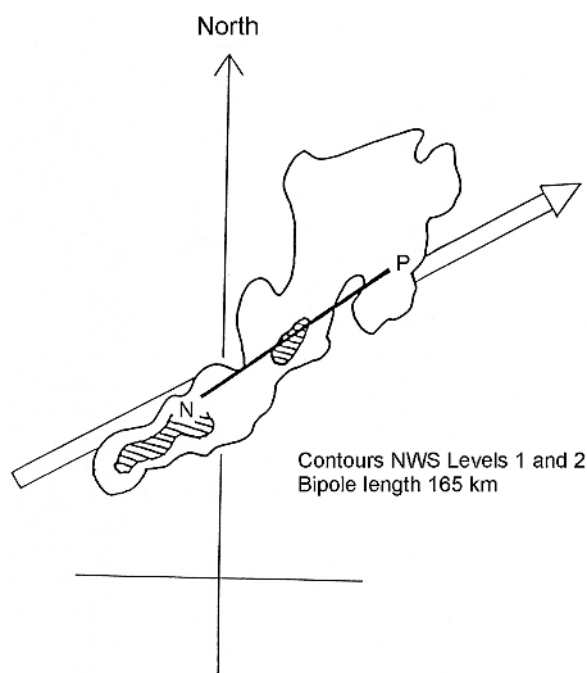


Fig. 2.6. An idealized lightning bipole is shown by the slanted solid line joining the geographic negative and positive centers N and P. The large outlined arrow indicates the geostrophic wind direction at all levels aloft. The contours represent two levels of radar reflectivity in imitation of the National Weather Service radar summary chart. Adapted from Engholm *et al.* (1990).

lightning flash rate was highest around the period of most intense convective rainfall. The positive lightning activity was confined mainly to the trailing stratiform region, and there was a correlation between the positive lightning

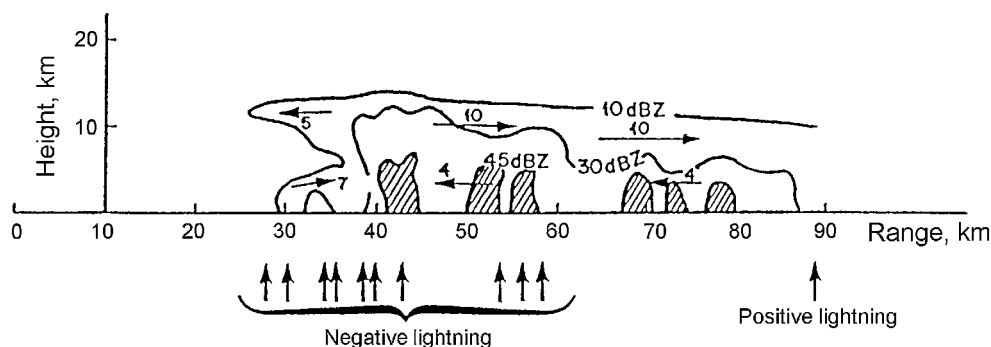


Fig. 2.7. Radar range height indicator (RHI) scan roughly aligned with the upper level geostrophic wind for the thunderstorm cluster on 28–29 June 1986 in Alabama. Doppler radial winds and cloud-to-ground lightning locations are also shown; the wind speeds in m s^{-1} are shown alongside the arrows. Adapted from Engholm *et al.* (1990).

flash rate and the area-integrated stratiform precipitation. Rutledge and MacGorman concluded from their analysis that the occurrence of positive lightning in the trailing stratiform region was a result of the rearward advection of positively charged ice particles from the upper levels of the convective cells. This charge advection mechanism may be viewed as an extension of the tilted dipole idea, proposed by Brook *et al.* (1982) as the mechanism for positive lightning production in winter storms in Japan (Chapter 8), from the convective scale (about 10 km) to the mesoscale (a few hundred kilometers). However, the charging of hydrometeors within the stratiform region was also viewed by Rutledge and MacGorman as a possibility. Similar bipolar patterns in both mid-latitude and tropical MCSs were found by Stolzenburg (1990), Engholm *et al.* (1990), Rutledge *et al.* (1990), Schuur *et al.* (1991), Hunter *et al.* (1992), Petersen and Rutledge (1992), and Rutledge and Petersen (1994). Figure 2.6, taken from Engholm *et al.* (1990), illustrates, for a generalized bipole in winter (January–March), the relative orientation of the bipole, the wind direction, and the precipitation structure. Figure 2.7, from the same study, presents (i) a vertical radar reflectivity profile, (ii) the Doppler radar winds, and (iii) the locations of positive and negative cloud-to-ground flashes for one thunderstorm cluster in summer. Engholm *et al.* (1990) concluded that lightning bipoles are aligned with the geostrophic wind (e.g., Houghton 1997) in winter storms and with the vertical wind shear (whose direction may be considerably different from that of the geostrophic wind) in summer storms, with the positive pole downwind (downshear) from the negative pole, the latter being located in proximity to the deepest convection. Hunter *et al.* (1992, Fig. 11) observed both of these types of bipole occurring simultaneously in summer at a distance of less than 100 km from each other. They and Marshall and Rust (1993) pointed out the lack of understanding of the processes leading to the apparently different types of bipole. Rutledge and Petersen (1994) showed that the number of ground flashes in the stratiform region is

strongly correlated with the vertical radar reflectivity profile and view this as an evidence of *in situ* charging in the stratiform region. However, they stated that the charge advection mechanism cannot be ruled out. Stolzenburg *et al.* (1994) concluded that no single mechanism seems likely to account for the four or more significant charge layers inferred from the vertical electric field profiles observed in stratiform clouds (Section 3.3). Schuur and Rutledge (2000a, b), from observations and modeling of one MCS, found that noninductive collisional graupel–ice charging (subsection 3.2.6) could account for as much as 70 percent of the total charge in the stratiform region, the remaining 30 percent being contributed by charge advection from the convective region. More information about the electrical charge structure of and electrification processes in stratiform clouds is found in Section 3.3. Hunter *et al.* (1992) and Marshall and Rust (1993) considered the possibility that positive lightning could originate in the convective region, propagate horizontally into the stratiform region, and produce a channel to ground there.

Holle *et al.* (1994), using the seven-station NSSL DF lightning locating network (Section 17.3), studied the cloud-to-ground lightning flash characteristics in a series of four MCSs that occurred in Oklahoma and Kansas on 3–4 June 1985. A detailed description of the life cycle of one of the MCSs, lasting about 11 hours, was given, as outlined next. The peak lightning flash rate was reached a few hours after storm initiation. As the storm matured, the fraction of positive flashes increased. During the decaying stage, negative lightning flash rates rapidly decreased. At most, a few percent of the flashes in convective regions lowered positive charge to ground, and positive lightning flash rates in convective regions rapidly decreased during the decaying stage, a trend similar to that observed for negative lightning flash rates. Positive lightning flash rates in the stratiform region, however, tended to increase until early in the decaying stage, positive flashes exceeding 60 percent of all the flashes in the stratiform area. Throughout the lifetime of the storm

there were more flashes per convective area than per stratiform area and more negative than positive flashes per area.

Nielsen *et al.* (1994), using lightning data from the NSSL network along with radar, satellite, and sounding data, examined the life cycle of a distinct mesoscale segment of the 10–11 June 1985 MCS. Positive flashes were dominant when the MCS was first developing. Negative flashes then became dominant during a period of intense leading-line convective activity with high storm tops. Finally, a period of relatively frequent positive flashes within the trailing stratiform region occurred during the demise of the MCS, after the vertical extent of the leading convective line decreased rapidly and markedly (30–40 dBZ echoes occurring mostly below the freezing level).

It is worth noting that not all MCSs show a bipolar pattern. Further, the stratiform region is not the only region of an MCS in which positive flashes can dominate ground flash activity. Several studies in the Great Plains of the United States (Rutledge *et al.* 1990; Schuur *et al.* 1991; Holle *et al.* 1994) have indicated that a substantial fraction, sometimes even a majority, of ground flashes within the convective region of an MCS can have positive polarity. More information on positive lightning occurrence is found in Chapters 5 and 8.

2.2.8. Lightning and rainfall

Thunderstorms typically produce both lightning flashes and rainfall, so the two quantities are expected to be related. Many researchers have attempted to quantify the relationship between lightning and surface rainfall. Battan (1965), using data from a rain gauge network and visual observations of ground flashes for 52 storms in Arizona, estimated an average of roughly 3×10^7 kg (3×10^4 m³) of rainfall per ground flash with a range of values for individual storms from 3×10^6 to 3×10^8 kg (3×10^3 to 3×10^5 m³) per flash. Williams *et al.* (1992), using radar and a lightning locating system in northern Australia, found a range from 2×10^9 to 3×10^{10} kg (2×10^6 to 3×10^7 m³) per ground flash for monsoon storms and a range from 9×10^7 to 10^9 kg (9×10^4 to 10^6 m³) per ground flash for continental storms. The average values were about 8×10^9 kg (8×10^6 m³) and 5×10^8 kg (5×10^5 m³) per flash for monsoon and continental storms, respectively. These and many other estimates are summarized in Table 7.6 of MacGorman and Rust (1998), the overall range of values extending over four orders of magnitude, from 3×10^6 to 3×10^{10} kg (3×10^3 to 3×10^7 m³) per ground flash. MacGorman and Rust (1998) also noted that in all the studies the largest observed rainfall yields per flash occurred with the smallest number of ground flashes per storm, typically less than 10–30. Holle *et al.* (1994) estimated 1.3 ground flashes per 10^6 m³ (10^9 kg) of rain in a US Great Plains MCS, comparable to the maximum

value of one ground flash per 10^6 m³ (10^9 kg) found by Williams *et al.* (1992) for continental storms in northern Australia. However, Piepgrass *et al.* (1982) found only about 10^4 m³ (10^7 kg) of rain per ground flash in Florida. Soula *et al.* (1998) reported 3×10^4 m³ (3×10^7 kg) per flash, and Molinie *et al.* (1999) reported a range from 3.5×10^3 to 47×10^3 m³ (3.5×10^6 to 47×10^6 kg) per flash in the Pyrenees. In isolated thunderstorms in Florida, Piepgrass *et al.* (1982) found that peak rainfall occurred less than 10 min after the peak flash rate, whereas the rainfall of squall lines often peaked 1 to 2 hr after the time of the peak ground flash rate (e.g., Rutledge and MacGorman 1988; Holle *et al.* 1994; Kane 1993b).

The relation between lightning and rainfall is characterized by large variability and is poorly understood. It apparently depends on location, storm type, stage of storm development, and other meteorological factors. For example, for coastal storms the direction of the prevailing winds interacting with the sea breeze along the coast (Lopez *et al.* 1991) is likely to influence this relation. Tapia *et al.* (1998), who analyzed 22 Florida thunderstorms, found that the convective rainfall mass per cloud-to-ground flash varied with the convective regime. There are lightning-producing storms with little or no surface rainfall (e.g., Reap 1986; Watson *et al.* 1994a), probably because much of the precipitation has evaporated before reaching ground (e.g., Colson 1960), and there are deep convection clouds, in particular marine cumulonimbi, producing rain but little or no lightning (e.g., Williams *et al.* 1992; Zipser 1994). Zipser (1994), who studied thunderstorm frequency over the oceans using observations from research ships, reported that in regions such as West Africa and South Asia the seasonal rainfall peak is actually accompanied by a thunderstorm minimum. He observed many tropical heavy-rain events that produced no lightning, even when their associated cloud tops reached 17 km, and hypothesized that most storms formed over oceans have updrafts weaker than a presumed threshold value (discussed below) required for the operation of the graupel–ice mechanism of cloud electrification discussed in subsection 3.2.6. As indicated above, Williams *et al.* (1992), in the Darwin area of Australia (12° S latitude), found that considerably more precipitation fell per ground flash from storms imbedded in monsoonal convection (monsoon storms) than from storms during the period typified by continental convection (continental storms). This latter observation appears to be consistent with Zipser’s (1994) finding that clouds formed over the ocean produce less lightning than clouds formed over the land.

According to Zipser and Lutz (1994), lightning is absent or highly unlikely if the vertical updraft speed does not exceed a threshold of roughly 6–7 m s^{−1} (mean) or 10–12 m s^{−1} (peak), regardless of cloud depth. Further, Petersen *et al.* (1996) attributed the lack of lightning activity

in oceanic convection to the fact that most of the hydrometeor mass was located below the freezing level, while the cloud tops were often observed to be higher than 15 km (subsection 2.2.6). However, if natural ground radioactivity or corona from ground plays a role in cloud electrification, as postulated by the convection cloud-electrification mechanism (subsection 3.2.6), then the lack of those phenomena over the oceans may explain the observed low electrical activity of the oceanic clouds even when their depth would indicate that they should yield a lightning output similar to that of the same-depth clouds formed over land. According to Boccippio *et al.* (2000a), who used OTD and lightning imaging sensor (LIS) data (subsection 2.5.3 and section 17.8), the bulk of the observed order-of-magnitude differences between land and ocean regional flash rates are accounted for by differences in storm spacing (density) and/or frequency of occurrence, rather than by differences in the storm instantaneous flash rates. Boccippio *et al.* (2000a) found that the latter rates only varied by a factor 2 on average.

The relationship between lightning and rainfall is also discussed in subsection 2.2.7 in considering the lightning bipolar pattern and in subsection 2.8.4 in considering the characteristics of lightning as a function of topography. Additional information on the lightning–rainfall relation is found in the works of Shih (1988), Cheze and Sanvageot (1997), Sheridan *et al.* (1997), Petersen and Rutledge (1998), Tapia *et al.* (1998), Carey and Rutledge (2000), Grecu *et al.* (2000) and Nesbitt *et al.* (2000), and in Section 7.19 of MacGorman and Rust (1998).

Concluding Section 2.2, we note that characteristics of individual storms in relation to season, location, and storm type are discussed in Section 2.8. Additional information on the electrical and other characteristics of thunderstorms is found, for example, in the book by MacGorman and Rust (1998).

In Sections 2.3 through 2.5, three characteristics of lightning activity, the annual number of thunderstorm days T_D , the annual number of thunderstorm hours T_H , and the annual ground flash density N_g , will be discussed. These three quantities are used to characterize the average local lightning activity rather than to characterize individual storms or parts of storm systems.

2.3. Thunderstorm days

The annual number of thunderstorm days T_D , also called the keraunic level, is the only parameter related to lightning incidence for which worldwide data are available extending over many decades. In fact, in some specific locations such data are available for more than a century. A thunderstorm day is defined as a local calendar day during which thunder is heard at least once at a given location. The practical range of audibility of thunder is about 15 km, the

maximum range of audibility being typically about 25 km (Section 11.2). Thunderstorm-day data are recorded at most weather stations by human observers, and apparently the first global climatological survey of thunderstorm occurrence based on T_D data was published by Klossovsky (1892) for 439 weather stations. The spatial distribution of long-term average values of T_D is often presented in the form of isokeraunic maps. Examples of such maps are shown in Figs. 2.8 and 2.9, for the entire world and for the United States, respectively.

Considerable effort has been expended to relate the annual number of thunderstorm days T_D to the ground flash density in $\text{km}^{-2} \text{yr}^{-1}$, N_g . Prentice (1977), for example, reviewed 17 proposed relations between these two parameters. Most of the relations are of the form

$$N_g = a T_D^b \quad (2.3)$$

where a and b are empirical constants. Apparently, the most reliable expression is the one proposed by Anderson *et al.* (1984a),

$$N_g = 0.04 T_D^{1.25} \quad (2.4)$$

This expression is based on the regression equation relating the logarithm of the five-year-average value of N_g measured with CIGRE 10 kHz lightning flash counters (subsection 2.5.1) at 62 locations in South Africa and the logarithm of the value of T_D as reported by the corresponding weather stations. The range for T_D was from 4 to 80, the range for N_g was from about 0.2 to about $13 \text{ km}^{-2} \text{yr}^{-1}$, and the correlation coefficient between the logarithms of N_g and T_D was 0.85. A similar equation, $N_g = 0.036 T_D^{1.3}$, based on the use of lightning flash counters in the USSR to determine N_g , was proposed earlier by Kolokolov and Pavlova (1972).

Equation 2.4 has been adopted by the Institute of Electrical and Electronics Engineers (IEEE) (Whitehead *et al.* 1993) and by CIGRE (the French acronym that stands for Conférence Internationale des Grands Réseaux Électriques à Haute Tension, or, in English, the International Conference on Large High Voltage Electric Systems) (CIGRE Document 63 1991) for estimating the lightning performance of power transmission lines. It is important to note that there is a considerable scatter of data points about the regression line corresponding to Eq. 2.4, as illustrated in Fig. 2.10. It is also worth noting that the application of Eq. 2.4, obtained for South Africa, to other areas of the world requires caution since the relation between N_g and T_D may well vary from one region to another (Prentice 1977; Rakov *et al.* 1990b). As noted above, however, this equation is very similar to the expression obtained in the USSR by Kolokolov and Pavlova (1972).

A characteristic of thunderstorm activity closely related to T_D is the number of thunder events, defined as isolated periods of thunder activity heard at a given weather station (e.g., Changnon 1988b, 1993). The number of thunder

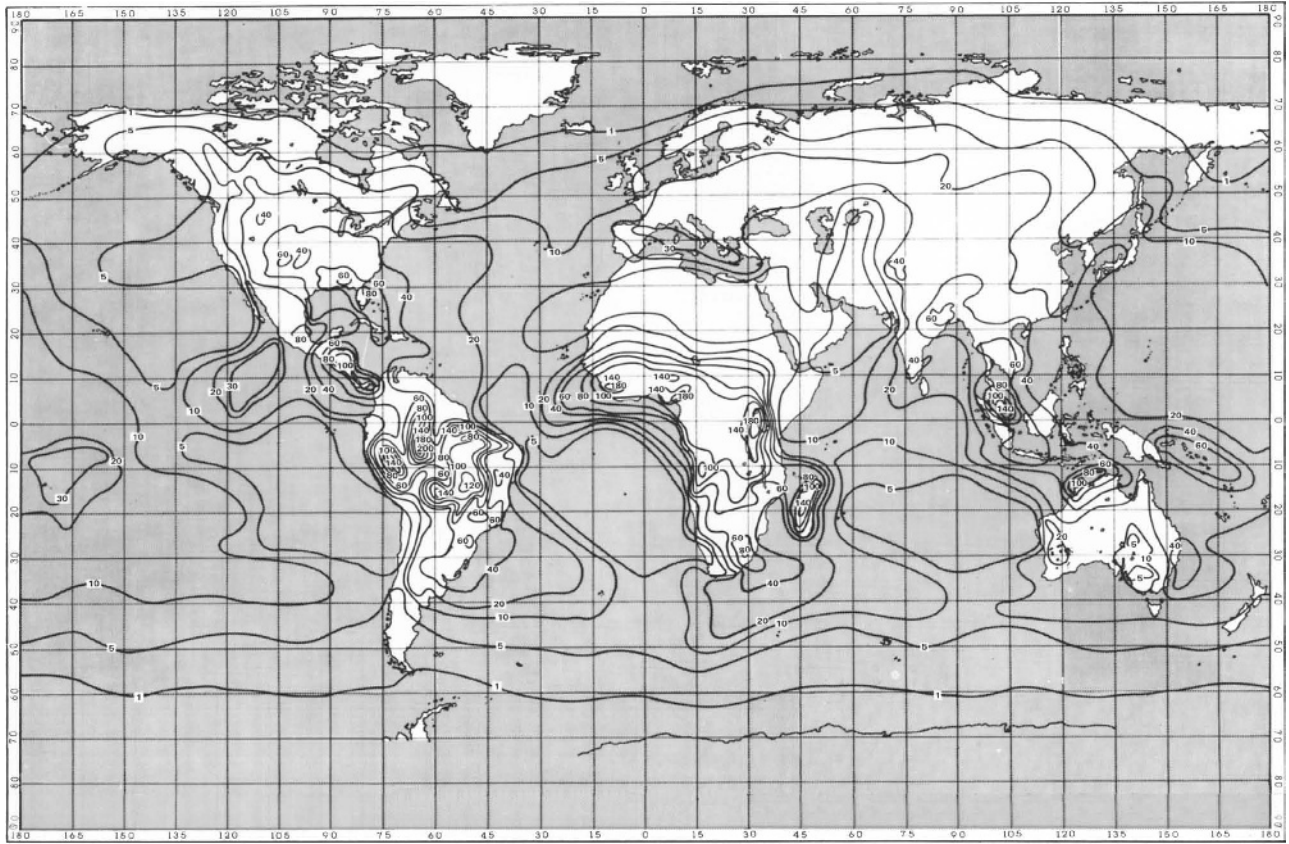


Fig. 2.8. A world map of mean annual number of thunderstorms days. Adapted from WMO Publication 21 (1956).

events can be viewed as the number of thunderstorms that occurred within the observation area of a weather station. A map of the average number of thunder events for the contiguous United States is considered in subsection 2.8.2 in relation to the spatial variation of lightning activity.

2.4. Thunderstorm hours

The annual number of thunderstorm hours T_H is a parameter that is potentially more closely related to the lightning incidence than T_D . Clearly, T_D does not distinguish between a small thunderstorm producing a few lightning flashes in tens of minutes and a large storm lasting for several hours and producing hundreds of flashes, while T_H does. As for T_D , T_H is routinely recorded by human observers at many weather stations, although some stations record only T_D . The frequency distributions of thunderstorm durations for 30 years at 36 stations in the United States were analyzed by Robinson and Easterling (1988). Kolokolov and Pavlova (1972), using weather station reports and lightning flash counter registrations (subsection 2.5.1) in the plain terrain of the European part of the USSR, found a linear relationship between N_g and T_H : $N_g = 0.05T_H$. MacGorman *et al.* (1984), using ground flash data from DF systems

(Section 17.3) operated in Florida and Oklahoma, inferred values of T_D and T_H that a human observer would report if located at grid points with 25 km spacing and compared the resulting correlations between N_g and T_D and between N_g and T_H . In each location, the correlation with T_H was better than with T_D . MacGorman *et al.* (1984) constructed an expression for the combined Florida and Oklahoma data,

$$N_g = 0.054T_H^{1.1} \quad (2.5)$$

where as before N_g is measured in $\text{km}^{-2} \text{ yr}^{-1}$. They used this to convert a map for T_H for the contiguous United States into a corresponding map for N_g . Changnon (1993), using 1986–9 data from three DF networks, found the ground flash density in Florida and Oklahoma to be in general agreement with the map of MacGorman *et al.*, while over most of the nation, particularly in the west, the predictions of Eq. 2.5 were a factor 2 higher. Nevertheless, Eq. 2.5 is similar to that previously reported by Kolokolov and Pavlova (1972) and to another relationship, $N_g = 0.04T_H$, obtained by Eriksson (1987) for the Transvaal Highveld region of South Africa. Note that the exponent on the right-hand side of Eq. 2.5 is close to unity and that in the expressions published by Kolokolov and Pavlova (1972) and by Eriksson (1987) it

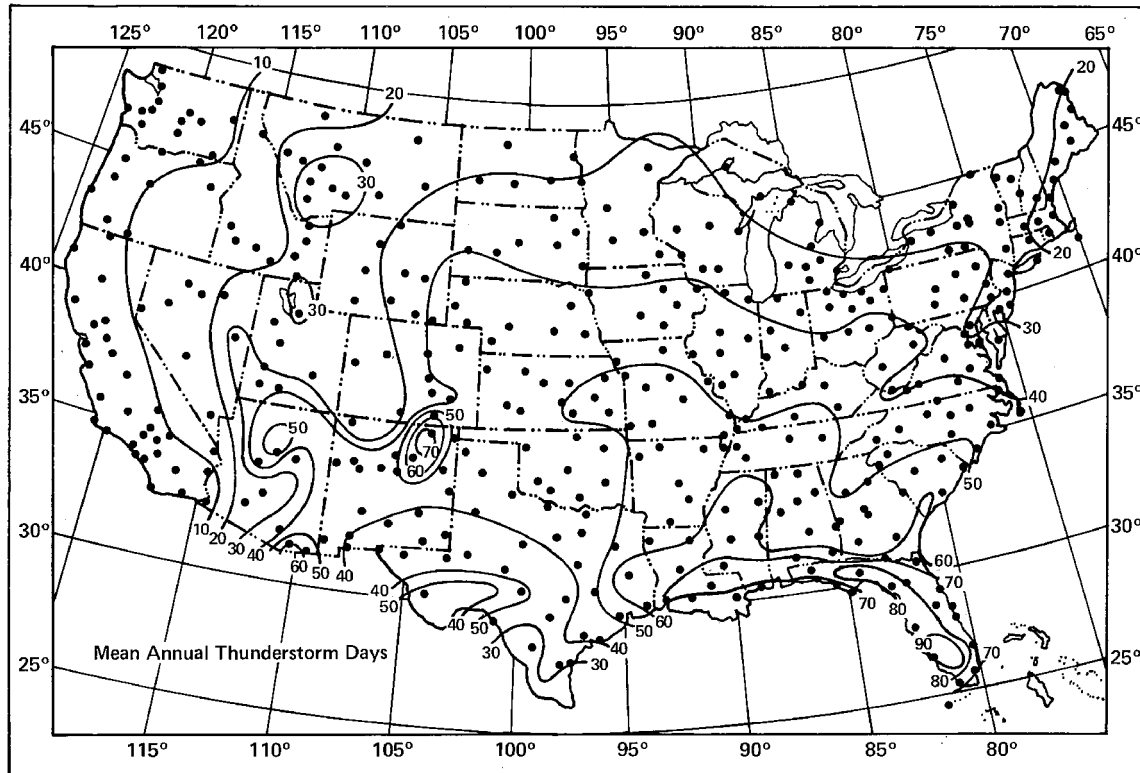


Fig. 2.9. A map of the mean annual number of thunderstorm days for the United States based on data from 450 air weather stations shown as dots (Changery, 1981). Most stations had 30-year records and all had at least 10-year records. Adapted from MacGorman *et al.* (1984).

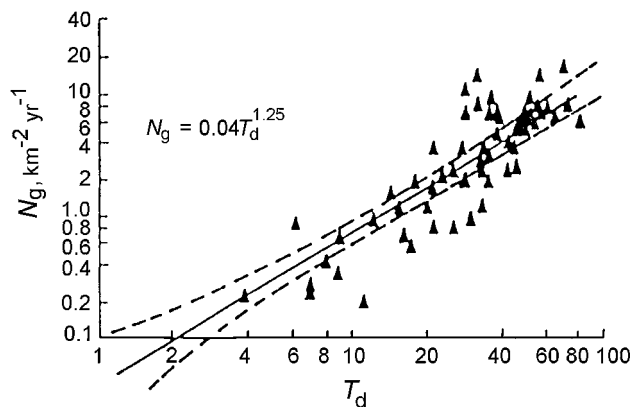


Fig. 2.10. Relationship between annual ground flash density and annual number of thunderstorm days. Adapted from Anderson *et al.* (1984a).

is unity; that is, N_g is linearly or nearly linearly related to T_H . Dulzon and Rakov (1991) suggested that T_H may not be better than T_D as a predictor of N_g . They found that the long-term annual number of lightning-caused outages of power lines that had similar geometrical and electrical characteristics and that were located in areas with different long-term values of T_D and T_H did not show a better correlation with

T_H than with T_D . Perhaps T_H records are more likely to suffer from human error than T_D records. Indeed, in the case of T_D it is just “yes” or “no” for each day, whereas in the case of T_H the observer must record the beginning and the end of storm, the end usually being defined as the time after which thunder is not heard for at least 15 minutes (e.g., Changnon 1988a). If thunder is heard again after, say, 20 minutes, a new beginning is recorded. It is possible that an observer may be hesitant to regard a given thunder as the final one when non-audible (out of thunder-hearing range) but visible lightning activity is present, a situation that is more likely to occur at night (e.g., Reap and Orville 1990).

2.5. Lightning flash density

Lightning flash density is often viewed as the primary descriptor of lightning incidence, at least in lightning protection studies. Flash density has been estimated from records of (i) lightning flash counters and (ii) lightning locating systems (Chapter 17) and can potentially be estimated from records of satellite-based optical or radio-frequency radiation detectors (Section 17.8).

About 50 percent of ground flashes in New Mexico and Florida strike ground at more than one point (Kitagawa *et al.* 1962; Rakov and Uman 1990b), the spatial separation

between individual channel terminations in Florida ranging from 0.3 to 7.3 km as determined using multiple-station TV records (Thottappillil *et al.* 1992). Most measurements of lightning flash density do not account for multiple channel terminations on ground. If all strike points separated by distances of some hundreds of meters or more are of interest, as is the case where lightning damage is concerned, measured values of ground flash density should, in general, be increased. The average number of ground terminations per flash in Florida thunderstorms, based on data reviewed by Rakov *et al.* (1994), is 1.7. Similar values were reported for New Mexico thunderstorms by Kitagawa *et al.* (1962). Thus, at least for these two locations, the correction factor for measured values of N_g to account for multiple channel terminations on ground should be about 1.7.

2.5.1. Lightning flash counters

The lightning flash counter (LFC) is an antenna-based instrument that produces a registration if the electric (or magnetic) field generated by lightning, after being appropriately filtered (the center frequency is typically in the range from hundreds of hertz to tens of kilohertz), exceeds a fixed threshold level. The output of an LFC is the number of lightning events and/or time sequence of lightning events recorded at a given location. The first antenna-based device that was used for the detection of lightning activity was the so-called *grozootmetchik* (Russian for thunderstorm detector), invented in 1895 by Popov (e.g., Popov 1896). Any LFC will record both ground and cloud lightning discharges, although only ground discharges are needed for the estimation of N_g . In order to “exclude” the undesired (in estimating N_g) cloud discharges from the total number of LFC registrations K , a correction factor Y_g , discussed later, is applied as a multiplier to K . Further, the product $Y_g K$ is converted to N_g by dividing it by the effective operating area πR_g^2 of the counter, where R_g is the effective radius, or range, for ground discharge detection, so that $N_g = Y_g K / (\pi R_g^2)$. Since the electric or magnetic field intensity (in the LFC frequency range) at a fixed distance from the lightning channel varies by two orders of magnitude or so (primarily owing to variations in the return-stroke current), any LFC with a fixed trigger threshold level will fail to register small events nearby while registering large events far away. Accordingly, the effective range for the counter is defined so that the “misses” within the effective range are exactly compensated by the “hits” outside it. An equivalent definition of effective range is the distance within which the number of flashes actually occurring equals the number registered by the counter. For a given LFC, the effective range depends on the field amplitude distribution at the source and may depend on field propagation conditions. In using an LFC, it is assumed that the lightning activity is reasonably homogeneous over its operating range, which is

several times greater than the effective range. About 40 percent of ground flashes are registered at a distance equal to the effective range, and lower percentages are registered beyond that range for a typical LFC (Bunn 1968; Prentice and Mackerras 1969; Anderson *et al.* 1979). Many years of measurement are required to obtain a meaningful average flash density from records of lightning flash counters (Section 2.6).

The correction factor Y_g , the fraction of ground flashes in the total number of LFC registrations K , is a function of the ratio z of the actual cloud and ground flash densities (Section 2.7) and of the ratio of the effective ranges, necessarily different, for cloud (R_c) and ground (R_g) discharges. The ratio of the effective ranges depends on the filter and triggering threshold level used. Thus the characteristics needed for proper interpretation of the registrations of any LFC, which are to be determined from independent measurements, include z , R_c , and R_g . Evaluation of these characteristics is not a trivial task since it requires the identification and ranging of both ground and cloud flashes within some tens of kilometers. Additionally, due to their dependence on the source, Y_g , R_c , and R_g can potentially change from one location to another, as discussed by, for example, Rakov (1986) and Rakov and Dulzon (1988).

The two most widely used counters are endorsed by CIGRE (see Section 2.3) and are called the CIGRE 500-Hz counter (Pierce 1956; Barham 1965; Golde 1966; Prentice and Mackerras 1969; Prentice 1972; Barham and Mackerras 1972; Prentice *et al.* 1975; Anderson 1977; Anderson *et al.* 1979) and the CIGRE 10-kHz counter (Anderson *et al.* 1973, 1979; Anderson 1977). The frequency designation refers to the center frequency passed by the filter that precedes the trigger circuit. Another “organization-recommended” counter, called the CCIR (International Radio Consultative Committee) or WMO (World Meteorological Organization) counter (Horner 1960), has not attained widespread use. In contrast with most other LFCs, the CCIR counter was not intended to differentiate between ground and cloud discharges. Its filter is centered at 10 kHz and is similar to that of the CIGRE 10 kHz counter; however, the threshold levels for these two counters differ considerably: 3 V m^{-1} for the CCIR (Horner 1960) and about 20 V m^{-1} for the CIGRE (Anderson *et al.* 1979). The CIGRE 10 kHz counter, employing a 10 kHz filter in conjunction with a 20 V m^{-1} trigger threshold, allows a lower percentage of cloud flashes in the overall counts than does the CIGRE 500 Hz counter, employing a 500 Hz filter in conjunction with a 5 V m^{-1} trigger threshold (Anderson *et al.* 1979). The effective ranges of the CIGRE 500-Hz counter were determined experimentally in Australia to be about 30 km for ground flashes and about 20 km for cloud flashes (Prentice and Mackerras 1969) and in South Africa to be about 37 km for ground flashes and

about 17 km for cloud flashes, about 20 percent of the total registrations being due to cloud flashes (Anderson 1980). The effective ranges of the CIGRE 10 kHz counter in South Africa were found to be about 20 km for ground flashes and about 6 km for cloud flashes, about 5 percent of the total registrations being due to cloud flashes (Anderson *et al.* 1979; Anderson 1980). Apparently, the effective range of the CCIR counter was not been properly determined by measurements, the reported values varying considerably (Horner 1960; Müller-Hillebrand 1963). Cooray (1986) estimated theoretically the effective range of this LFC to be about 140 km. Some comments on the general theory and the inherent weaknesses of LFCs are given by Brook and Kitagawa (1960) and by Bunn (1968), and the operation of the two CIGRE counters and the CCIR counter is discussed in detail by Cooray (1986). If Y_g and R_g are known, LFCs can provide reasonably accurate data on ground flash density. We consider some of these data next.

Prentice (1977) summarized much of the published and unpublished data on average flash density that had been obtained using LFCs, visual observations, and instruments that measure electric field change. The most extensive flash counter studies come from Scandinavia (e.g., Müller-Hillebrand *et al.* 1965), Australia (e.g., Mackerras 1978), and South Africa (e.g., Anderson 1980; Anderson *et al.* 1984a, b). Southeast Queensland, Australia, has a total flash density of $5 \text{ km}^{-2} \text{ yr}^{-1}$, of which $1.2 \text{ km}^{-2} \text{ yr}^{-1}$ are ground flashes; Norway, Sweden, and Finland have measured ground flash densities between 0.2 and $3 \text{ km}^{-2} \text{ yr}^{-1}$ depending on location, and South Africa has ground flash densities from below 0.1 to about 12 or $13 \text{ km}^{-2} \text{ yr}^{-1}$, depending on location. Probably the most reliable ground flash density map based of LFC observations is that compiled for South Africa, the observation period covering 11 years (CIGRE Document 63 1991). Ground flash density maps based on more recent LFC observations are presented by Sunoto (1985) for Java, Indonesia, by Romualdo *et al.* (1989) for Mexico, by Diniz *et al.* (1996) for Minas Gerais, Brazil, and by Rakov *et al.* (1990a, b) for the Tomsk region of Russia. Appreciable differences between the spatial distributions (maps) of N_g and T_D are typically noticed. In the study of Rakov *et al.* (1990a, b), the Monte-Carlo technique was used to combine long-term (24-year, on average) statistical distributions of T_D from weather stations and the distributions of N_g per thunderstorm day measured over five years with a network of LFCs.

Crozier *et al.* (1988), using a single-station magnetic direction finding system (essentially an LFC) with a nominal range of 180 km located in Southern Ontario, Canada, estimated ground flash densities of about 1.6 and $2.4 \text{ km}^{-2} \text{ yr}^{-1}$ for 1982 and 1983, respectively. Additional information from the system's output included the lightning

flash rates, the percentage of positive flashes, the percentage of multiple-stroke flashes, interstroke intervals, flash duration, and relative stroke amplitudes within the flash.

An attempt to design an LFC capable of recording ground and cloud discharges separately with the same effective range was made by Mackerras (1985). His device, dubbed CGR1, processed electric field changes using various criteria to distinguish the signals produced by ground lightning, by cloud lightning, and by non-lightning sources. First, it was assumed that a 400 V m^{-1} threshold for the overall electric field change assures an effective range of about 14 km for both ground and cloud discharges. Further, the occurrence of lightning must be confirmed either by the detection of “at least one fast positive-going change greater than 40 V m^{-1} or by the detection of a burst of RF noise in a band centered at about 150 kHz, with a detection bandwidth of about 40 kHz”. It appears that the detection of cloud flashes largely relies on the RF noise criterion. Since this noise is not always detectable, cloud flashes can be preferentially rejected by the device. Finally, if a signal is recognized as due to lightning, the following three quantities are compared: (i) the overall field change, (ii) the largest fast positive-going field change, and (iii) the sum of fast positive-going field changes. Depending on the results of the comparison (the quantitative criteria as well as the gain of the RF noise channel were set by trial and error), the event is classified as either a ground flash or a cloud flash.

An advanced version of this LFC, dubbed CGR3, has five registers for recording (i) cloud flashes, (ii) positive cloud-to-ground flashes, (iii) negative cloud-to-ground flashes, (iv) so-called distant flashes, and (v) so-called overload conditions (Mackerras and Darveniza 1994). For CGR3, different effective ranges are specified for cloud flashes (12.2 km), for positive ground flashes (16.2 km), and for negative ground flashes (13.6 to 15.4 km, depending on latitude). It is possible that some parameters of the flash identification algorithm, which was apparently adjusted for thunderstorms in Brisbane, Australia are inadequate for other geographical locations. Therefore, the results of the studies based on the use of CGR instruments in various geographical locations, presented in Sections 2.7 and 2.8, should be viewed with caution.

2.5.2. Lightning locating systems

Locating lightning discharges with reasonable accuracy requires the use of multiple-station networks. The principles of operation of multiple-station lightning locating systems (LLS) are described in Chapter 17. In this chapter we will consider only LLSs that employ the radio-frequency electromagnetic fields produced by lightning to determine one location (usually the ground strike point) per stroke or flash, as opposed to systems that are designed to

image lightning channels, largely inside the cloud. Ground-strike-point locating systems are presently used in many countries to acquire lightning data that can be used for mapping N_g . Any such LLS fails to detect relatively small cloud-to-ground flashes (particularly near the periphery of the network) and fails to discriminate against some cloud flashes, unwanted in determining N_g . The corresponding system characteristics, the detection efficiency, and the selectivity with respect to ground flashes are influenced by network configuration, position of the lightning relative to the network, the system's sensor gain and trigger threshold, sensor waveform selection criteria, lightning parameters, and field propagation conditions. The interpretation of system output in terms of N_g is subject to a number of uncertainties (e.g., Lopez *et al.* 1992), but multiple-station lightning locating networks are by far the best available tool for mapping N_g . Difficulties with data interpretation may arise because of the temporal and spatial inhomogeneity of the networks that from time to time undergo upgrading, expansion, adjustment, or reconfiguration. For example, according to Wacker and Orville (1996, 1999a), the 1994 upgrade of the NLDN to employ both magnetic direction finding and time-of-arrival sensors (Section 17.5) resulted in a significant decrease in the measured mean peak current of both positive and negative strokes and a doubling of the annual positive flash count. Huffines and Orville (1999) reported that the number of events identified by the NLDN as positive flashes in 1995–6, after the upgrade, was a factor 2–4 larger than before the upgrade. Wacker and Orville (1999b) attributed the observed decrease in mean peak currents and increase in flash count to the reduction during the 1994 upgrade of the minimum electromagnetic pulse width required in order for an event to be accepted by the system as a stroke to ground. More details on the 1994 upgrade (which was completed in 1995) are found in Section 17.5.

The present NLDN employs 106 sensors (Fig. 17.6), and different versions of the network have provided lightning data covering the continental United States since 1989. From 1989 to 1995, the time and location of flashes have been archived along with estimates of the peak current of the first stroke, the polarity of the first stroke, and the number of strokes in the flash (the so-called “multiplicity”). Since 1 January 1995, the archived data have included the time, location, peak current, and polarity of both first and subsequent strokes detected by the NLDN. The different methods of grouping individually measured strokes into flashes before and after the 1994 upgrade are described in Section 17.5. The ground flash density derived from the locating system's data is computed by dividing the region of interest into small rectangles and accumulating the total number of flashes occurring in each rectangular “grid cell” over the time interval of interest. The gridded data are

smoothed before converting them to flash density contours (Byerley *et al.* 1995).

Orville (1991) published the first N_g map for the contiguous United States based on the data (over 13.4 million ground flashes) from the 1989 NLDN, composed of 114 DF sensors. He assumed a detection efficiency of 70 percent throughout the network (accordingly, the measured numbers were multiplied by 1.4) and used a grid size of 30 by 50 km. A 10-year (1989–98) N_g map for the contiguous United States is shown in Fig. 2.11. As seen in this figure, the variation of N_g from one region to another is more than two orders of magnitude, from $0.1 \text{ km}^{-2} \text{ yr}^{-1}$ or less on the Pacific Coast to $14 \text{ km}^{-2} \text{ yr}^{-1}$ or more in Florida. Maps for 1989–91 for the United States were published by Orville (1994), for 1992–5 by Orville and Silver (1997), for 1989–96 by Huffines and Orville (1999), and for 1995–7 by Orville and Huffines (1999). According to Huffines and Orville (1999), about 25 million cloud-to-ground flashes occur annually over the contiguous United States. Both annual and monthly N_g maps for Florida, based on the 1986–95 data, are found in the work of Hodanish *et al.* (1997). Values of N_g based on data accumulated over a few years can be affected significantly by the occurrence of just one storm (Orville and Songster 1987; Orville 1991).

The availability of flash characteristics along with locations allows ground flash density studies that were not possible before the implementation of the NLDN and similar networks. For example, Reap and MacGorman (1989) showed the first maps of ground flash density for negative flashes and for positive flashes using data from the NSSL lightning locating network (Section 17.3). Orville (1994) plotted N_g for positive flashes as well as contours of the percentage of positive flashes for the United States for 1989 through 1991. It is even possible to plot N_g for flashes having strokes with inferred peak currents (Section 17.5) greater than, say, 100 kA (e.g., Lyons *et al.* 1998). Interestingly, lightning locating system data indicate that measured flash densities can vary by an order of magnitude over distances of 20–30 km, partly owing to meteorological effects such as those occurring along the Florida coastline (Maier *et al.* 1979). Similar observations have been made by Lopez and Holle (1986) for lightning in central Florida and in northeastern Colorado.

Krider (1988), using a three-station DF lightning locating network, studied the distances between successive flashes in three small thunderstorms in Florida. He found that the mean distance between successive flashes ranged from 3.2 to 4.2 km, and the maximum distance ranged from 10 to 12 km. Lopez and Holle (1999), using NLDN data, found that, in storms of small and medium dimensions in Florida and Colorado, the median distance between successive flashes was approximately 4 to 5 km, while for a large mesoscale storm in Oklahoma it was about 9 km.

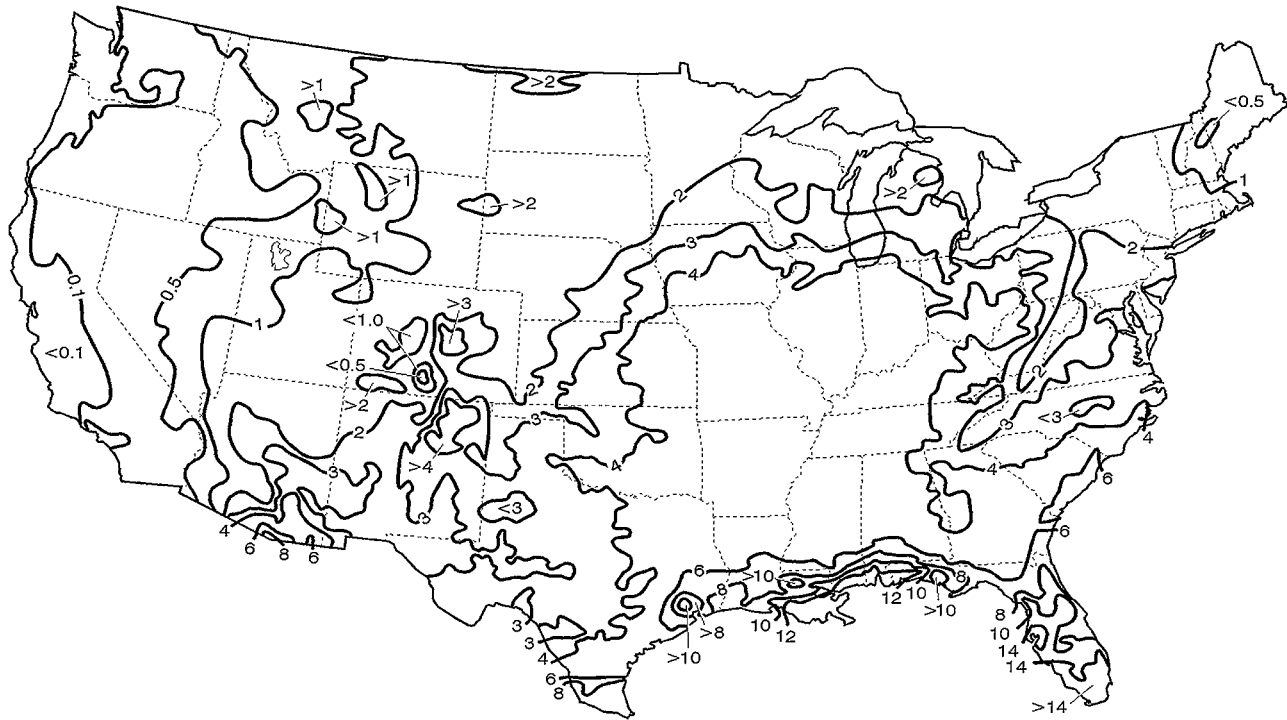


Fig. 2.11. Annual ground flash density map for the United States based on 1989–98 NLDN data. Courtesy of Global Atmospheric, Inc.

2.5.3. Satellite-based detectors

With the advent of Earth-orbiting satellites, it has become possible to map systematically the worldwide lightning activity by detecting the optical or radio-frequency radiation emitted by cloud and ground discharges. It is worth noting that in order to obtain N_g maps from satellite observations, a spatial distribution of the fraction of discharges to ground relative to the total number of lightning discharges, discussed in Section 2.7, is needed. Apparently, the first reports of lightning detected with an instrument flown on a satellite were published by Vorpahl *et al.* (1970), who used an optical detector on the Orbiting Solar Observatory satellite OSO-2. Satellites in low Earth orbit spend a relatively short time over any given storm and, hence, record only a small fraction of the discharges. In the case of optical detection before 1995, lightning flashes were visible only in photographs taken during night-time passes, and the detection efficiency was often less than two percent (Christian *et al.* 1989). In the case of radio-frequency detection, which is independent of the time of day, a significant fraction of the data can be contaminated by interference from man-made sources on the ground (e.g., Kotaki *et al.* 1981a, b; Kotaki and Katoh 1983). More information on satellite-based lightning detectors is found in Section 17.8.

Although the best spatial resolution for optical detectors in orbit before 1995 was of the order of 100 km, it was possible, nevertheless, to estimate total flash densities

(both cloud and cloud-to-ground discharges combined) and to determine ratios of activity in different geographical locations and in different seasons. It was not, however, possible to distinguish between cloud and cloud-to-ground discharges. Orville and Henderson (1986), using photographic data from a Defense Meteorological Satellite Program (DMSP) satellite, studied the global distribution of lightning at midnight between 60° S and 60° N for 365 consecutive days (from 1 September 1977 to 31 August 1978). They gave a series of lightning maps for each month and for groups of months in which the locations of the discharges were well correlated with known features of the general atmospheric circulation. These maps show the concentration of lightning over continents and islands as opposed to oceans, the monthly ratio of flash density over land to flash density over oceans ranging from 5.3 to 10 (7.7 on average), which is consistent with the surface hourly observations from research ships in the tropics reported by Zipser (1994). Possible explanations for the lack of lightning over the oceans are discussed in subsection 2.2.8. Orville and Spencer (1979), using dusk and midnight photographic data obtained in 1974 and 1975 with two DMSP satellites, estimated that the global flash rate was 123 s^{-1} at dusk and 96 s^{-1} at midnight, with a potential error of about a factor 2. They reported that there was 1.4 times more lightning during summer in the northern hemisphere than during the southern summer. Turman and Edgar (1982), using DMSP

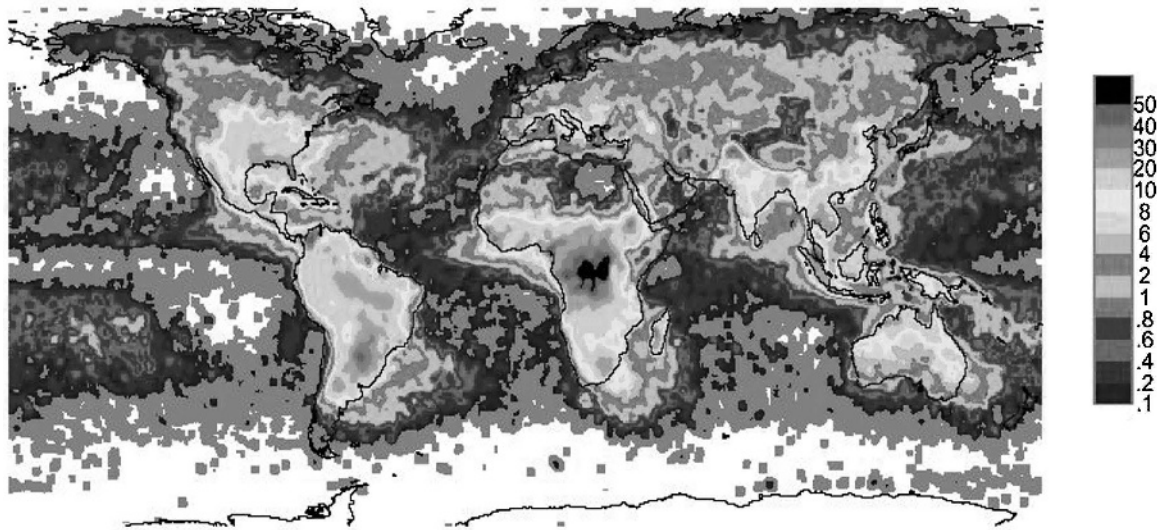


Fig. 2.12. See also the colour plate section. A global map of total lightning flash density in $\text{km}^{-2} \text{yr}^{-1}$ based on data from two satellite detectors, OTD (five years) and LIS (three years). Grey areas correspond to a flash density range $0.01\text{--}0.1 \text{ km}^{-2} \text{yr}^{-1}$ and white areas to a flash density $<0.01 \text{ km}^{-2} \text{yr}^{-1}$. Courtesy of H.J. Christian, NASA/Marshall Space Flight Center.

satellite data, estimated a global flash rate of 40 to 120 s^{-1} with a seasonal variation of about 10 percent. Thirty-seven percent of the global lightning activity originated over the ocean at dawn and 15 percent at dusk. Turman (1978) using an array of 12 photodiodes (each having a field of view of 700 km by 700 km) flown on a DMSP satellite, observed 10 000 flashes from 24 storm complexes during 15 orbits in September 1974 and March 1975. The average flash density within the sensor field of view was $6 \times 10^{-8} \text{ km}^{-2} \text{s}^{-1}$, which, if extrapolated to the whole Earth, represents a global flash rate of about 30 s^{-1} . Kotaki *et al.* (1981a, b) and Kotaki and Katoh (1983) detected lightning with the Japanese Ionosphere Sounding Satellite-b (ISS-b), which sensed HF radiation at 2.5, 5, 10, and 25 MHz. They presented worldwide lightning maps for a 2 yr period. They found the global lightning frequency to be 64 s^{-1} for the northern hemisphere spring, 55 s^{-1} for the summer, 80 s^{-1} for the fall, and 54 s^{-1} for the winter. The characteristics of early satellite lightning experiments are summarized by Christian *et al.* (1989, Table 1).

A significant improvement in the optical observation of lightning from space was achieved with the launch, on 3 April 1995, of NASA's Optical Transient Detector (OTD) (Christian *et al.* 1996a). The OTD, also described in Section 17.8, detects lightning by looking for small transient changes in light intensity during both day-time and night-time. An OTD flash is a collection of optical pulse groups (adjacent pixel illuminations within the same 2 ms frame) that are geographically adjacent and that have no "dead time" between groups greater than 333 ms. OTD's spatial errors are, on average, about 20–40 km, and its temporal errors are less than 100 ms (Boccippio *et al.*

2000b). Data averaging over 55 or 110 days is required to remove the diurnal lightning cycle bias. Like previous satellite-based optical detectors, the OTD cannot distinguish between ground and cloud discharges. The global lightning distributions (e.g., Christian and Latham 1998) obtained are similar to those based on earlier satellite studies. Boccippio *et al.* (2000b) compared lightning detected by the OTD and by the NLDN. They found that the OTD detection efficiency for ground flashes was about 46 to 69 percent and is likely to be slightly higher for cloud flashes. Data from the OTD are found on the website <http://thunder.msfc.nasa.gov/otd.html>. The OTD stopped sending data in April 2000. A follow-on lightning detector, called the Lightning Imaging Sensor (LIS), was launched on 27 November 1997 as part of the Tropical Rainfall Measuring Mission (TRMM). More information on the LIS is found in Section 17.8. Thomas *et al.* (2000) compared lightning detected by the LIS and by a three-dimensional time-of-arrival lightning imaging system, as discussed in Section 17.8. They found that lightning discharges confined below 7 km altitude were less likely to be detected than discharges extending to higher altitudes. Data from the LIS are found on the website <http://thunder.msfc.nasa.gov/lis.html>. A total lightning flash density map based on data from two satellite detectors, OTD (five years) and LIS (three years), is shown in Fig. 2.12. The data are cross-normalized, and detection efficiency adjustments are applied as a function of detector, local hour, and, for the OTD, sensor threshold setting. An additional correction is applied to the OTD data in the so-called South Atlantic Anomaly. The average values of the detection efficiency estimates are 49 percent for OTD and 82 percent for LIS. The data represent a

0.5×0.5 degree composite with a 2.5 degree spatial-moving-average operator applied. The satellite data shown in Fig. 2.12 yield a global flash rate of 45 s^{-1} with an estimated uncertainty of $\pm 5 \text{ s}^{-1}$.

As noted in Section 2.1, Brooks (1925) estimated that the global flash rate was about 100 s^{-1} on the basis of data obtained by human observers. The satellite data are in remarkably good agreement with this value, considering the assumptions that were necessarily made both in analyzing the satellite data and in determining Brook's value.

2.6. Long-term variations in lightning incidence

Since lightning activity varies from year to year, obtaining average values of T_H , T_D , or N_g potentially requires observations over many years, although it is conceivable that stable long-term average values do not exist. Anderson *et al.* (1979) recommended taking data for at least 11 years, one solar cycle, although Freier (1978) found little evidence for coupling between solar activity and thunderstorms. Kitagawa (1989) studied the number of thunderstorm days per month for a period of 100 years in Japan both in winter (November, December, January, and February) over the Japan Sea coast and in summer (July and August) over three different regions, inland plains, inland mountains, and open sea coast. Variations in the number of thunderstorm days over the 100-year period for two locations showing different trends are given in Fig. 2.13.

It has been suggested (e.g., Williams 1992, 1994) that global lightning activity is a sensitive function of the global surface air temperature. Jayaratne (1993), using a CGR3 lightning flash counter (subsection 2.5.1), observed that the monthly lightning activity in Gaborone, Botswana, in summer increased by an order of magnitude for a 2°C rise in wet bulb temperature. The latter quantity represents the effects of both temperature and humidity and is described, for example, by Williams (1995). Price and Rind (1994a), using cloud top height as a proxy for lightning frequency, estimated an approximate 5–6 percent change in global lightning frequencies for every 1°C global warming or cooling, while Reeve and Tuomi (1999), who used data from the OTD (subsection 2.5.3), estimated a 40 percent change in lightning activity for a one-degree temperature change. Michalon *et al.* (1999) calculated from a cloud electrification and lightning model that a surface warming of 2°C would produce a 10 percent increase in the annual global flash frequency. According to Williams (1992), single-station Schumann resonance measurements that represent ELF (3 Hz to 3 kHz) standing waves in the Earth-ionosphere cavity, due to lightning from many storms occurring worldwide, can be used for the estimation of global lightning activity, as discussed in subsection 13.4.3. If the Earth's climate warms owing to an increase in trace gas concentrations in the atmosphere, a temperature increase of

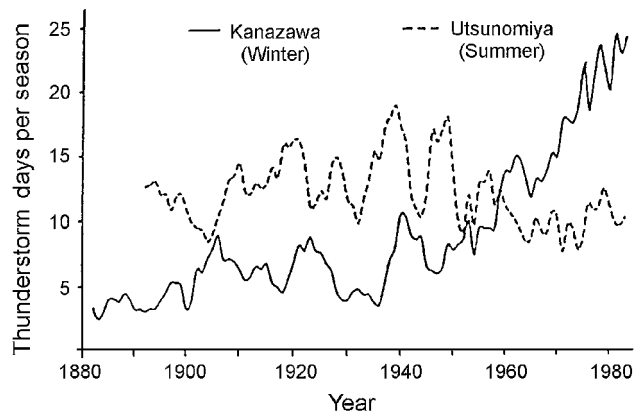


Fig. 2.13. Long-term variation in the number of thunderstorm days per season for winter storms along the Japan Sea coast (Kanazawa) and for summer storms over the inland Kanto Plain (Utsunomiya). Adapted from Kitagawa (1989).

1.5 to 4.5°C being predicted by the middle of the twenty-first century (Intergovernmental Panel on Climate Change 1990), significant increases in lightning activity should be expected. Price (2000) showed that upper-tropospheric water-vapor variability and global lightning activity are closely linked. Small changes in upper-tropospheric water vapor have a large impact on the greenhouse effect on Earth.

Dulzon (1992) presented T_H data from a weather station in Siberia for a 50-year period between 1936 and 1985. It can be inferred from these data that the mean annual number of thunderstorm hours determined for moving 20-year periods first increased to a maximum value of about 50 percent more than the 50-year mean and then decreased to a minimum value of about 40 percent less than the 50-year mean. According to Changnon (1985), in North America the continental averages of frequency of thunderstorm days exhibited an average increase of 15 percent from 1901 to the 1930s and then a decrease, interrupted by a peak in the 1970s, of about 10 percent from 1940 to 1980. He also reported that northern ($\geq 50^\circ \text{N}$) stations in Europe, Asia, and North America all exhibit upward trends from 1901 to 1980. Mackerras (1977, Fig. 5) showed a decreasing tendency of annual thunderstorm days in Brisbane, Australia, during the 1911–68 period. Anderson *et al.* (1984a) reported that the annual values of N_g measured using the CIGRE 500 Hz LFC in Pretoria (South Africa) in 1970–1 and in 1972–3 were more than 50 percent in excess of the long-term (15-year) mean value.

Byerley *et al.* (1995) showed N_g maps based on NLDN data for 1992 and 1993 and noted the significant variability in flash density for these two years. In 1992, only 16 million flashes were detected, whereas over 24 million were detected in 1993. The NLDN was stable over this

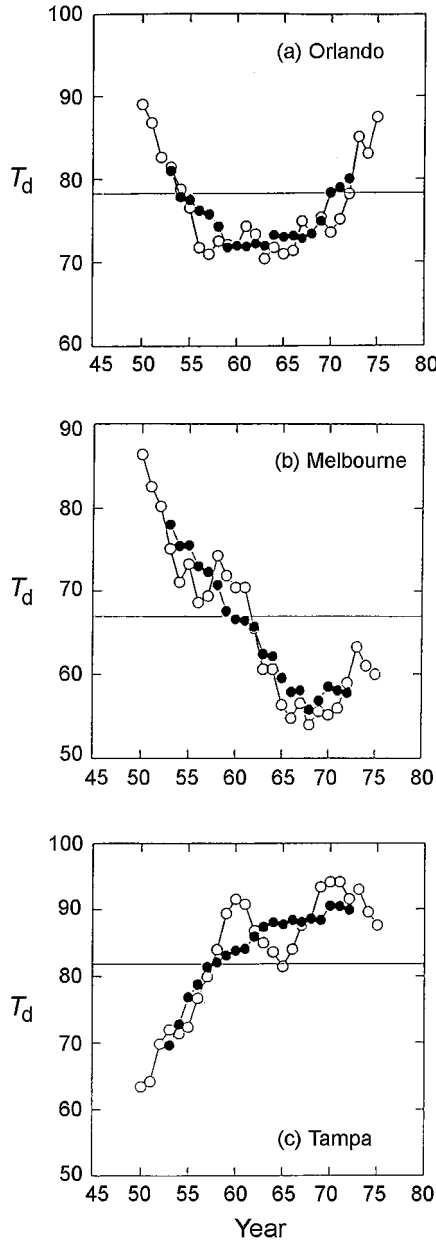


Fig. 2.14. Variation in the annual number of thunderstorm days over a period of approximately 30 years (1948 to 1977) at three Florida weather stations: (a) Orlando, (b) Melbourne, and (c) Tampa. ○, five-year moving average; ●, 11-year moving average; —, 30-year mean. Adapted from Uman *et al.* (1994).

two-year period, in terms of instrumentation and data-processing algorithms.

Figure 2.14 shows the variation in the annual number of thunderstorm days T_D over a period of approximately 30 years (1948 to 1977) at three Florida weather stations: Orlando (Fig. 2.14a), Melbourne (Fig. 2.14b) and Tampa (Fig. 2.14c). To reduce the effect of the year-to-year scatter, running averages with five-year and 11-year windows were

performed on the data. The 30-year mean value is indicated by a horizontal line on each of these figures. The corresponding data (not given here) for the annual number of thunderstorm hours T_H show more variation relative to the long-term mean than do the T_D data. Note that, for the same period of time, trends at different stations are different. Some of these variations might be due to changes in the local observation conditions, except for the variations reported by the weather station in Tampa, which has a high-quality thunderstorm record over the period from 1896 to 1995 (Changnon 2001a).

2.7. Ratio of cloud flashes to cloud-to-ground flashes

The literature on the ratio of cloud to cloud-to-ground flashes has been reviewed by Prentice and Mackerras (1977), Livingston and Krider (1978), Rakov and Dulzon (1984), Mackerras and Darveniza (1994), and Mackerras *et al.* (1998). We first present earlier studies that indicated that there is a latitudinal dependence of the ratio, a result that is widely accepted but apparently not confirmed by more recent, although not necessarily more accurate, measurements. Overall, the ratio z of cloud to cloud-to-ground flashes can vary significantly depending on storm type and other factors, the average value being probably about 3.

Prentice and Mackerras (1977) found, from an examination of the literature, that the ratio had mean values of 5.7 for latitudes between 2° and 19° , 3.6 between 27° and 37° , 2.9 between 43° and 50° , and 1.8 between 52° and 69° . They fitted 29 observations with the equation

$$z = N_c/N_g = 4.16 + 2.16 \cos 3\lambda$$

where N_c is the cloud flash density and λ is the latitude. All the experimental data on z fall within a factor 3 of the value given by the above equation, most within a factor 2. Mackerras (1985) employed an automatic device (CGR1; subsection 2.5.1) designed to distinguish between the electric fields of ground and cloud discharges to determine the ratio of cloud flashes to ground flashes for the period September 1982 to May 1984 in Brisbane, Australia ($\lambda = 27.5^\circ$ S). The ratio for about 6100 total flashes was about 3 with a range from 0.9 to 24.7 for individual days having at least 100 total flashes. The ground flash density was $1.2 \text{ km}^{-2} \text{ yr}^{-1}$ and the cloud flash density was $3.7 \text{ km}^{-2} \text{ yr}^{-1}$, in agreement with the LFC data reported by Prentice (1977) (subsection 2.5.1).

Previously, Pierce (1970) had suggested the use of the following expression to describe the fraction of total flashes that involve ground:

$$p = N_g/(N_c + N_g) = 0.1 + 0.25 \sin \lambda$$

This equation provides the ratios $z = (1 - p)/p = 9$ at the equator and $z = 1.8$ at the poles. Rakov and Dulzon (1984)

argued that, because of the large scatter in the data, the use of trigonometric functions is not justified and gave a linear regression equation relating z and λ . Price and Rind (1993) found that the value of z in a thunderstorm increases with increasing thickness of the cold cloud region (0°C level to cloud top). They obtained data on ground flashes from the BLMDLF lightning locating network (Section 17.3) and used the radar-determined convective cloud top heights to infer total flash rates. The relation between total flash rate and convective cloud top height is discussed in subsection 2.2.6 and illustrated in Fig. 2.4. Although the relation of z to the cloud thickness above the 0°C isotherm deduced by Price and Rind (1993) seems to be due in part to their representation of the total flash rate as a function of cloud top height, it apparently explains why the observed values of z are larger in tropical regions than in the mid-latitudes. A similar explanation was previously offered by Rakov and Dulzon (1984) from a comparison of the observed values of z in different parts of the world with typical cloud vertical dimensions and typical isotherm heights as a function of latitude.

Mackerras and Darveniza (1994) reported measurements of z using the CGR3 instrument (subsection 2.5.1) at 14 locations whose latitudes varied from 60°N to 27°S . These measurements, made from 1987 through 1991, the period of observation ranging from two months to five years, in effect refute the hypothesis of the strong latitudinal dependence of z . In contrast to the earlier studies, many of which involved visual observations, with the CGR3 Mackerras and Darveniza found tropical values of z , in the range 0.5 to 3.4 (with a weighted mean of 2.3 for the latitude range 0° to 20°), similar to the values found in subtropical and temperate regions (where the range is from 1.1 to 3.8 and the weighted mean is 2.2 for the latitude range 20° to 40°). It is only at high latitudes (40° to 60°) that their measured values of z (with range 1.0 to 1.5 and weighted mean 1.3) agree with the earlier studies. The stated uncertainty in z -values is about ± 50 percent. No dependence on site altitude, which varied from 0 to 2800 m, was observed. Mackerras and Darveniza estimated that the global mean value of z based on measurements at 14 sites is 1.9, the values at 13 of the sites being within a factor 2 of the global mean. Mackerras *et al.* (1998) reported that the estimates of z presented by Mackerras and Darveniza are underestimates because the assumed value for the effective range for cloud discharges was too high. The adjusted estimates of z given by Mackerras *et al.* from CGR3 registrations at 11 sites are 4.0 for the latitude range 0° to 20° , 3.2 for the latitude range 20° to 40° , and 1.9 for the latitude range 40° to 60° . As noted in subsection 2.5.1, the performance characteristics of the CGR3 instruments estimated in Brisbane, Australia, may be inapplicable to other regions, even after the introduction of various adjustment factors. Therefore,

the estimates of z given by Mackerras *et al.* (1998) as well as other results obtained using the CGR3 (e.g., Baral and Mackerras 1992, 1993; Mackerras and Darveniza 1994; Jayaratne *et al.* 1995; Levin *et al.* 1996; Yair *et al.* 1998; Jayaratne and Ramachandran 1998) should be viewed with caution. Thus, none of the proposed latitudinal dependencies of z can be accepted with confidence.

In the following, we will consider various factors that influence the z -value at a given location. From field-mill records for summer thunderstorms in Florida ($\lambda = 28^\circ\text{N}$) Livingston and Krider (1978) found that between 42 and 52 percent of all lightning discharges were cloud-to-ground discharges during the “active storm period” (subsection 2.2.1) and that during the final storm period only about 20 percent were to ground. For five storms on three days, 43 percent of 552 total flashes involved ground. Holzer (1953) also noted that the percentage of cloud flashes increases in the latter stage of a storm.

Rutledge *et al.* (1992) found that the ratio of the cloud flash rate to the cloud-to-ground flash rate was correlated with the total flash rate for MCSs in the Darwin, Australia, area. The ratio varied from 2.7 to 17 as the total flash rate changed from 1 to 40 min^{-1} . Apparently the higher total flash rates were primarily a consequence of increased intracloud discharge activity, consistent with the finding of Lhermitte and Krehbiel (1979) in Florida. However, Jayaratne *et al.* (1995), using the CGR3 in Gaborone, Botswana, found that the ratio of cloud flashes to ground flashes decreased with increasing total flash rate, implying a larger proportion of ground flashes in the more active thunderstorms. The ratio was generally highest in the early and later stages of individual storms, falling to its lowest values during the more active intermediate stages. The apparent discrepancy between the observations of Rutledge *et al.* (1992) in Australia and Jayaratne *et al.* (1995) in Botswana is likely to be due to differences in the methods used for the identification of different types of lightning. Yair *et al.* (1998), also using the CGR3, observed that the monthly average value of z for the Tel Aviv, Israel, area varied significantly from month to month in a given season and from year to year (1989–6) for a given month. The range of variation was more than an order of magnitude, from less than 0.5 to 5. For MCSs in the United States, Mazur and Rust (1983) and Schuur *et al.* (1991) found z to be 40 and 8, respectively. Lang *et al.* (2000) reported on two intense convective storms in Colorado that were characterized by very low cloud-to-ground flash rates ($< 1\text{ min}^{-1}$) while exhibiting relatively high cloud flash rates ($> 39\text{ min}^{-1}$), so that the value of z was greater than 30. They related this high value of z to strong updrafts. Rakov and Dulzon (1986) reported a significantly larger value of z for frontal storms, 4.5, than for air-mass storms, 1.8, in the Tomsk region of Russia (see subsection 2.8.5 below). Boccippio *et al.* (2001) studied

the distribution of z over the continental United States using OTD and NLDN data (Sections 17.8 and 17.5) covering a four-year period to estimate the total number of flashes and the number of cloud-to-ground flashes, respectively. They reported a range of variation in z from 1.0 over the Rocky and Appalachian mountains to 8–9 in the central-upper Great Plains, the average value being between 2.5 and 3. The range of variability at a given location was comparable to the range of values reported from the early studies for regions from the tropics to the mid-latitudes. Thus, the observed large scatter in z at a given latitude is probably due to the dependence of z on various other factors combined with the often limited sample size.

2.8. Characteristics of lightning as a function of season, location, and storm type

Many results in this section are given without attempting to speculate on the physical processes possibly responsible for the various observed features. We refrain from such speculations here because of the complexity and variability of lightning generation mechanisms, some reported results being potentially deficient in the identification of all significant factors involved and in the statistical significance of these results. Additionally, the interpretation of data is often complicated by the inherent limitations of lightning detection techniques, discussed in Chapter 17. Further information on lightning activity in relation to meteorological, geographical, and other factors can be found in Rakov (1992) and Holle and Lopez (1993).

2.8.1. Season

Orville and Songster (1987) and Orville *et al.* (1987), using the US East Coast DF lightning detection network (Section 17.3) in 1984–5, found that the percentage of positive lightning during the summer was less than 5 percent, but in October it began to increase, reaching a maximum of slightly over 80 percent in February and then decreasing to less than 10 percent in April. In the winter, most of the storms were dominated by positive flashes and occurred over the ocean. Orville and Huffines (1999), using 1995–7 NLDN data for the contiguous United States, reported that the minimum percentage of positive flashes occurred in the July–August period and the maximum occurred in January–February, as illustrated in Fig. 5.2. The percentage of positive flashes as a function of month from the studies of both Orville and Silver (1997) and Orville and Huffines (1999) is given in Table 5.2. Further, Orville and Songster (1987) and Orville *et al.* (1987) inferred that the median return-stroke peak current increases by approximately 50 percent in the winter season for both positive and negative flashes. Orville and Huffines (1999) reported, however, from 1995–7 NLDN data, that the median return-

stroke peak current for first strokes in negative flashes remained approximately constant from January through November at about 20 kA, increasing to 24 kA in December. The median positive current peak was 25 kA in February, 15 kA in July, and 24 kA in December. Note that the peak current values reported by Orville and Huffines (1999) are averages over the contiguous United States and that the seasonal dependence of peak current may vary from one region to another. Finally, Orville *et al.* (1987) observed that the percentage of single-stroke negative flashes increased from 40 percent in the summer to over 80 percent in the winter (along the US east coast from North Carolina to Maine), although both figures are likely to be overestimates owing to the fact that the locating system fails to detect small subsequent strokes (Rakov *et al.* 1994; Cummins *et al.* 1998).

Moore and Orville (1990), using ground flash data from the State University of New York at Albany (SUNYA) DF lightning detection network (Section 17.3) in conjunction with satellite, surface, upper air, and lake-temperature data, studied the characteristics of Great-Lakes-induced storms. These storms occur in the fall and winter and produce only a few cloud-to-ground flashes, mostly positive (except for one storm). Both the positive and negative flashes were associated with higher peak currents than in the summer. Also in winter, relatively high lightning activity has been found in the rainbands over the warm Gulf Stream off the east coast of the United States near North Carolina during northwesterly offshore flow (e.g., Biswas and Hobbs 1990; Orville 1990b, 1993; Dodge and Burpee 1993).

Brook (1992), using an electric field measuring system with a bandwidth from 0.1 Hz to about 1 MHz, observed that for negative lightning in winter (i) the initial breakdown pulses (Section 4.3) have larger amplitudes and (ii) the stepped leader durations (Section 4.4) are shorter than in summer lightning. No seasonal differences were found for positive lightning. On the basis of the observations for negative lightning, Brook (1992) concluded that winter discharges may be considerably more energetic than summer ones.

Hojo *et al.* (1989), using an LLP (Lightning Locating and Protection) DF lightning locating network, studied the seasonal variations of ground flash density, the distribution of peak magnetic field for first strokes, and the polarity of ground flashes in the coastal area of the Sea of Japan. Ground flash density maps for summer and winter were reported to be markedly different. In winter, ground strikes occurred primarily to the sea surface, and those to the land were no more than 30 km from the coast. For positive lightning, the peak magnetic field was larger in winter. However, for negative lightning no such seasonal variation was observed, the latter finding being in contrast to the results of Orville *et al.* (1987). The seasonal variation of the polarity

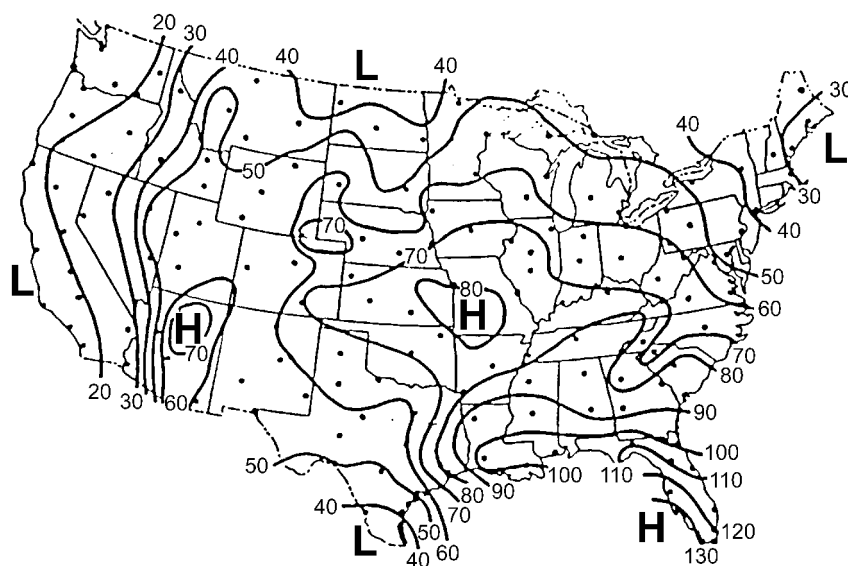


Fig. 2.15. Average annual numbers of thunder events in the contiguous United States. Adapted from Changnon (1988b).

of ground flashes in Japan is similar to that reported by Orville *et al.* (1987) for the northeastern part of the United States. The seasonal variations of the time characteristics of return-stroke field waveforms were analyzed by Ishii and Hojo (1989). They found that at distances of 100–300 km the zero-crossing time (subsection 4.6.3) for both polarities is shorter in winter. More information on winter lightning in Japan is found in Chapter 8.

2.8.2. Region

We discuss here the dependence of lightning characteristics on location in general. The dependences on latitude and topography are discussed separately in subsections 2.8.3 and 2.8.4, respectively.

Cooray and Jayaratne (1994) compared lightning characteristics determined from their electric field measurements in Sri Lanka with similar characteristics observed in Sweden (Cooray and Perez 1994) and in Florida (Rakov and Uman 1990a, b; Thottappillil *et al.* 1992). The geometric-mean interstroke intervals in Sri Lanka, Sweden, and Florida are 57, 48, and 57 ms, respectively. The mean number of strokes per flash and percentage of single-stroke flashes are 4.5 and 21 percent in Sri Lanka, 3.4 and 18 percent in Sweden, and 4.6 and 17 percent in Florida. The similarity of the lightning characteristics given above for the three different geographical regions is rather remarkable.

The literature on the height and magnitude of the negative cloud charge (subsections 3.2.2 and 3.2.3) and the electric dipole moment change per flash was reviewed by Jacobson and Krider (1976), and the literature on the flash duration and the frequency of occurrence of continuing current was reviewed by Livingston and Krider (1978).

Apparently, negative charge centers always occur at about the same cloud-temperature level, typically -10 to -34 °C, although these temperatures occur at different heights above the local terrain in different locations (Fig. 3.7).

Changnon (1988b), using thunderstorm records at 152 first-order weather stations distributed across the contiguous United States for a 30-year period (1948–77), studied “thunder events”, which were defined as periods of discrete thunder activity heard at a given weather station. In fact, a thunder event is the thunderstorm, the end of which is usually defined as the time after which thunder is not heard for at least 15 minutes (Section 2.4). The sum of the durations of all thunder events during a year is the annual number of thunderstorm hours, T_H . Since a thunderstorm day T_D (Section 2.3) comprises one or more thunder events (thunderstorms), the two characteristics are closely related, the correlation coefficients ranging, according to Changnon (1988b), from $+0.84$ to $+0.98$.

A map of the average annual number of thunder events is shown in Fig. 2.15. Annual averages of thunder events are high along the Gulf Coast (> 100), in the central United States (Kansas, Missouri, Illinois, with > 75 events), and in the southwest (Arizona, with 60 events). Thunder events are least numerous along the west coast (< 20) and in the northeast (< 30). A peak in thunder events is present in the central United States in all months, and its position is always closely related to the major center of cold frontal activity. The peak in the southwest is related to the summer monsoon intrusion of moist tropical Pacific air and related frontal activity. The peak along the Gulf Coast of Florida is a result of sea-breeze-induced convergence, localized heating, and occasional tropical disturbances in summer and fall. Maps of ground flash density for the contiguous

United States, based on data from the NLDN, are discussed in subsection 2.5.2. These maps show more than two orders of magnitude variability in N_g (see Fig. 2.11).

Westcott (1995), using NLDN data in and around 16 cities for June, July, and August of 1989–92, found an enhancement of 40 to 85 percent in the cloud-to-ground lightning activity over and downwind of many of these cities. Further, Orville *et al.* (2001), from a similar study for the period from 1989 through 2000, reported elevated ground flash densities, in both summer and winter, centered over and downwind of the metropolitan area of Houston, Texas. They suggested that the observed enhancement of lightning activity could be caused by enhanced convergence associated with the urban heat island and altered microphysical processes associated with anthropogenic pollution.

2.8.3. Latitude

Thomson (1980) reviewed the literature on inter-stroke time interval and number of strokes per flash and found no statistically significant correlation with latitude. This does not, however, preclude the latitude dependence which some investigators (Pierce 1970; Takeuti *et al.* 1975a) have inferred should exist, but which others (Harris and Salman 1972) have found not to exist. Rather, Thomson (1980) concluded that “the distributions of both interstroke intervals and the number of strokes per flash are more sensitive to local influences, including measurement and sampling techniques, than they are to latitude effects”.

Orville and Huffines (1999) reported that the mean number of strokes per negative flash (also called the “multiplicity” when it is reported by the NLDN) appeared to increase with decreasing latitude in the eastern half of the United States, from 1–1.5 in Maine to over 3 in Florida. As noted earlier, these multiplicity values are probably underestimates due to the NLDN’s failure to detect small strokes. Thus, the observed latitudinal trend could be due in part to the fact that both first and subsequent strokes in Florida are, on average, larger than those in higher latitudes, as indicated below for first strokes.

Rakov and Dulzon (1984, 1988) and Pinto *et al.* (1997), on the basis of their reviews of the literature on lightning current measurements in different countries, and Orville (1990a), on the basis of the analysis of current estimates provided by an LLP DF lightning locating network in the United States, argued that lightning peak currents for first strokes tend to increase as the latitude decreases. Some of the variation in peak current reported by Orville (1990a), however, might be due to instrumental or propagation effects.

A discussion of the possible latitudinal dependence of the ratio of cloud to cloud-to-ground flashes was given in Section 2.7.

2.8.4. Topography

Rakov *et al.* (1989) used a single-station lightning locating system “Ochag” (Russian for source location) in the North Caucasus region of Russia to estimate the spatial variation in N_g with topography. In this system, azimuth is determined via magnetic direction finding and distance is roughly estimated via the so-called EH ranging algorithm (Kononov *et al.* 1986; Kononov and Petrenko 1996). This algorithm utilizes the expected difference between the electric and magnetic field waveforms in the distance interval from 15 to 100 km. Rakov *et al.* (1989) found an average value of N_g that was a factor 1.7 higher for a mountainous area than for a plain-terrain area, the two areas being about equally covered by the lightning locating system. Orville (1994), however, reported lower values of N_g for the Appalachian Mountains than for the neighboring areas to the east and west, from the NLDN data for 1989–91. Kitagawa (1989) reported that the number of thunderstorm days per month (averaged over July and August) in Japan was highest over the inland Kanto Plain, lowest over the open sea coasts, and intermediate over the mountainous regions.

Reap (1986) analyzed cloud-to-ground lightning location data for the summers of 1983 and 1984 from the BLM network (Section 17.3) in the 11 western states of the contiguous United States. The lightning locations and occurrence rates were examined in relation to the topographic features of the generally mountainous western US terrain. Reap (1986) found a high correlation between terrain elevation, which ranged from sea level to near 3 km, and the time of maximum lightning activity: at the higher elevations, maximum lightning activity occurred in the early afternoon to midafternoon whereas at lower elevations it occurred later. The cloud-to-ground lightning activity was also found to increase with increasing terrain elevation, consistent with observations of Rakov *et al.* (1989) but in contrast with the results of Orville (1994) for the Appalachian Mountains.

King and Balling (1994), using 1989 and 1990 data from the BLM network, analyzed the diurnal variations of lightning flashes in Arizona during the summer monsoon season. In much of the state, the maximum occurred in the mid-to-late afternoon period, but it occurred closer to midnight in the large valley of central Arizona, a finding consistent with that of Reap (1986) discussed above.

Lopez and Holle (1986) used networks of LLP DFs in northeastern Colorado and in central Florida to study the diurnal and spatial variations of the lightning ground flash density (Section 2.5) and to relate those results to the geographic and climatic characteristics of the two regions. The geographic feature of primary importance in Colorado is the mountains and in Florida is the interface between the waters of the Gulf of Mexico or the Atlantic Ocean and the land of the peninsula. For the summer of 1983, detailed maps of the ground flash density as a function of

the time of day were given for both locations. Lopez and Holle (1986) showed from these maps that the temporal and spatial distributions of lightning are clearly related to the local topographic features. Reap (1994), using 1987–90 data from the SUNYA DF lightning locating network for Florida, observed organized coastal maxima in lightning activity related to land–sea-breeze convergence zones that form in direct response to the low-level wind flow, with two primary maxima in lightning activity near Tampa and west of Cape Canaveral and a minimum over Lake Okeechobee.

Lucas and Orville (1996), using lightning data from one LLP DF installed at Kavieng, Papua New Guinea, examined the frequency of cloud-to-ground lightning over the ocean as well as its diurnal variation. The data, acquired during 57 days in January and February 1993, indicated that flash counts over the land sector were approximately nine times higher than flash counts over the ocean sector. The highest lightning activity occurred around local midnight for both land and ocean sectors. Hidayat and Ishii (1998), using data from a four-station LLP DF–TOA network on the island of Java, Indonesia, reported the average annual ground flash density over the entire island to be more than an order of magnitude greater than over the Indian Ocean about 100 km south of the island. Further, Hidayat and Ishii (1999) estimated that the median current peak for lightning over the ocean was at least 20 percent higher than for lightning over the land.

Boccippio *et al.* (2000a) analyzed lightning data from the OTD and LIS (subsections 2.5.3 and 17.8) for variability between land and ocean, various geographic regions, and different convective “regimes”. They found that ocean storms produce lightning at rates that are not dissimilar from those of continental storms, as noted in subsection 2.2.1. According to Boccippio *et al.* (2000a), the bulk of the observed differences in regional flash rate between land and ocean is accounted for by differences in storm spacing (density) and/or frequency of occurrence.

Takeuti *et al.* (1975a) and Takeuti (1976), using a video camera and an electric field meter, found that cloud-to-sea lightning and cloud-to-ground lightning were similar in the number of strokes per flash and the interstroke interval distribution.

Orville and Huffines (1999) reported, from 1995–97 NLDN data, that median current peaks for first strokes in negative flashes were relatively high, typically exceeding 26 kA, along US continental coastal areas, particularly along the West Coast. Mountainous regions appeared to have lower median negative current peaks of about 15 to 20 kA. Median positive current peaks exceeded 40 kA in the upper midwest but were less than 10 kA in Louisiana and Florida, although the latter value is an underestimate due to misidentification and hence inclusion of some cloud flashes in the ground flash data.

2.8.5. Storm type

For frontal and for air-mass storms, Rakov and Dulzon (1986) compared the statistical distributions of (i) the ratio of cloud to cloud-to-ground discharges, z (Section 2.7), (ii) the return-stroke peak current, I , (iii) the number of strokes per flash, n , averaged over the storm, and (iv) the time interval between strokes, Δt . The distributions of z , I , and n were estimated from electric field measurements in the Tomsk region of Russia, while the distributions of Δt were those previously analyzed for latitudinal dependence by Thomson (1980). Rakov and Dulzon (1986) found that frontal storms are characterized by a larger mean value of z , 4.5 (13 storms) versus 1.8 (22 storms) for air-mass storms, a larger mean value of n , 2.7 (11 storms) versus 1.8 (14 storms), and a smaller geometric mean value of Δt , 51 ms versus 69 ms. The difference between the distributions of I was found to be statistically insignificant, the geometric mean values of I being 15 and 17 kA for frontal and air-mass storms, respectively.

Holzer (1953), working in New Mexico, found that at all stages of the development of an individual thunderstorm cell, in frontal storms ground flashes constituted a larger fraction of the total than in nonfrontal storms. In both classes of storms the cloud flashes became relatively more frequent in the latter stages of the life cycle of the cell. The average number of “repeated elements” (strokes) in cloud-to-ground flashes for air-mass storms was found to be about two, while for the frontal storms the average number was nearly five. The average intervals between strokes were similar, about 50 ms. Schonland (1956) also reported that in South Africa there was a greater number of strokes per flash in frontal storms than in air-mass storms. Kitterman (1980), using a streak camera, observed a greater percentage of “high-order multiple-stroke ground flashes” in frontal storms than in air-mass storms, while flash durations in these two types of storms were similar.

2.9. Lightning incidence to various objects

2.9.1. General information

We first describe briefly how cloud-to-ground lightning “decides” on its ground termination point. More details are given in Section 4.2 and subsection 18.3.2. Ground flashes are normally initiated by stepped leaders that originate in the thundercloud. As the downward-extending leader channel, usually negatively charged, approaches the ground, the enhanced electric field intensity at irregularities of the Earth’s surface or at protruding grounded objects increases and eventually exceeds the breakdown value of air. As a result, one or more upward-moving leaders are initiated from those points. When one of the upward-moving leaders from the ground contacts a branch of the downward-moving stepped leader, the point of lightning termination on ground is determined.

Grounded vertical objects produce relatively large electric field enhancement near their upper extremities so that upward-moving connecting leaders from these objects start earlier than from the surrounding ground and, therefore, serve to make the object a preferential lightning termination point. In general, the higher the object, the greater the field enhancement and hence the higher the probability that a stepped leader will terminate on the object. In the limit, when the height (the field enhancement capability, to be more exact) of the object becomes so large that the upward-moving leader from the object tip can be initiated by in-cloud charges or, more likely, by in-cloud discharge processes, as opposed to being initiated by the charge on the descending stepped leader, the object becomes capable of initiating upward lightning, as discussed in Chapter 6. The latter, as opposed to “normal”, downward lightning, would not occur if the object were not there. Ground-based objects with heights ranging from about 100 to 500 m experience both downward and upward flashes, the proportion being a function of object height. Eriksson (1987) derived the following equation for the annual lightning incidence N (in yr^{-1}) to ground-based objects, including both downward and upward (if any) flashes:

$$N = 24 \times 10^{-6} H_s^{2.05} N_g \quad (2.6)$$

where H_s is the object height in meters and N_g is the ground flash density in $\text{km}^{-2} \text{yr}^{-1}$. To do so, he employed (i) observations of the lightning incidence to structures of heights ranging from 20 to 540 m in different countries, (ii) the corresponding local values of the annual number of thunderstorm days T_D , and (iii) Eq. 2.4. It is worth noting that the majority of observed lightning strikes involved in the derivation of Eq. 2.6 correspond to taller structures and that the observed lightning incidence to smaller-height structures might have been affected by the presence of surrounding objects such as buildings and trees. As a result, Eq. 2.6 is probably less reliable for $H_s < 60$ m or so than for greater heights. Eriksson (1978) tabulated the observed percentage of upward flashes as a function of a free-standing structure's height, reproduced in Table 2.3. Eriksson and Meal (1984) fitted the data in Table 2.3 using the following expression:

$$P_u = 52.8 \ln H_s - 230 \quad (2.7)$$

where P_u is the percentage of upward flashes and H_s is the structure height in meters. This equation is valid only for structure heights ranging from 78 to 518 m, since $P_u = 0$ for $H_s = 78$ m and $P_u = 100$ percent for $H_s = 518$ m. Structures with heights less than 78 m are not covered by Eq. 2.7, because they are expected to be struck by downward flashes only, and structures with a height of greater than 518 m are not covered because they are expected to experience upward flashes only.

Table 2.3. *The percentage of upward flashes from tall structures. Adapted from Eriksson (1978)*

Reference	Structure height, m	Percentage of upward flashes
Pierce (1972)	150	23
	200	50
	300	80
	400	91
McCann (1944)	110	8
	180	24
	400	96
Berger (1972)	350 ^a	84
Gorin (1972); Gorin <i>et al.</i> (1976)	540	92 ^b
Garbagnati <i>et al.</i> (1974)	500 ^c	98

^a An effective height of 350 m was assigned by Eriksson to Berger's 70 m high mountain-top towers to account for the enhancement of the electric field by the mountain, whose top is 640 m above Lake Lugano (914 m above sea level). Pierce (1971) assigned a different effective height, 270 m, to Berger's towers (Section 6.1).

^b 50 percent of the flashes recorded in this study were classified as “unidentified”. The relative incidence of upward flashes is based upon analysis of only the identified data.

^c The towers of Garbagnati *et al.* were 40 m high, located on mountain tops 980 and 993 m above sea level (Berger and Garbagnati 1984). See Section 6.1.

In practice, as stated above, it is often assumed that structures having heights less than 100 m or so are struck by downward lightning only and that the upper height limit can be simply taken as 500 m. Accordingly, (i) the total lightning incidence N to a structure is the sum of the downward-flash incidence N_d and the upward-flash incidence N_u if the structure height is in the range from about 100 to 500 m, (ii) $N = N_d$ for structures shorter than 100 m, and (iii) $N = N_u$ for structures taller than 500 m. If both downward and upward flashes are expected, they are often treated separately in estimating the lightning incidence to an object, as described below.

2.9.2. Downward flashes

When the incidence of downward lightning is estimated, it is common to ascribe a so-called equivalent attractive (or exposure) area to the grounded object. The attractive area can be viewed as an area on flat ground surface that would receive the same number of lightning strikes in the absence of the object as does the object placed in the center of that area. In other words, in computing

lightning incidence to a structure, the structure is replaced by an equivalent area on ground. For a free-standing structure whose plan-view dimensions are much smaller than its height (such as a mast, tower, or chimney), this area, A , is taken as circular and is generally given by $A = \pi R_a^2$, where R_a is the equivalent attractive radius, discussed later. For straight, horizontally extended structures (such as power lines or their sections), the equivalent attractive area is rectangular and is sometimes termed the “shadow zone” or “attractive swath”. For example, if a power line has length l and effective width b (usually taken as the horizontal distance between overhead shield wires or between the outer phase conductors), its equivalent attractive area is generally estimated as $A = l(b + 2R_a)$, where R_a is the equivalent attractive distance, generally thought to be approximately equal to the equivalent attractive radius for a free-standing structure of the same height (Eriksson 1987; Rakov and Lutz 1990). Further, the local ground flash density N_g is assumed to be spatially uniform in the absence of the structure, so that the downward lightning incidence to the structure is found as

$$N_d = AN_g \quad (2.8)$$

Usually N_g is in $\text{km}^{-2} \text{yr}^{-1}$, so that A should be expressed in km^2 to obtain N_d in yr^{-1} (strikes per year).

The equivalent attractive radius (or distance) R_a is usually assumed to be a function of structure height H_s and is generally expressed as

$$R_a = \alpha H_s^\beta \quad (2.9)$$

where α and β are empirical constants. The procedures used to obtain Eq. 2.9 from data on lightning incidence to structures of different height are given, for example, by Eriksson (1978, 1987). In Eq. 2.9, both H_s and R_a are in meters, and various values of α and β have been proposed. For example, Whitehead *et al.* (1993) gave $\alpha = 2$ and $\beta = 1.09$ for transmission lines, while the CIGRE Document 63 (1991) recommended $\alpha = 14$ and $\beta = 0.6$ (see also subsection 18.4.3). The attractive radius for individual strikes should depend on the charge carried by the descending leader, this charge being correlated with the associated return-stroke peak current (Section 4.6 and subsection 18.3.2). In this regard, Eq. 2.9 should be understood as representing the entire distribution of peak currents. In the so-called electrogeometric approach (Section 18.3), which is widely used for the estimation of lightning incidence in lightning protection studies (e.g., CIGRE Document 63 1991), the equivalent attractive radius depends explicitly on the statistical distribution of lightning peak currents (e.g., Eriksson 1987; Rakov and Lutz 1988, 1990).

Estimation of N_d from Eq. 2.8 implies a reasonably long-term value of ground flash density and yields a long-term average value of lightning incidence. For example,

Table 2.4. The number of houses out of a total of 200 (the percentage, if divided by 2) expected to be struck by lightning n times over T years ($N_g = 4 \text{ km}^{-2} \text{yr}^{-1}$; $A = 1200 \text{ m}^2$)

Number of years of observation, T	Number of times struck by lightning, n				
	0	1	2	3	>3
10	191	9	0	0	0
20	182	17	1	0	0
30	173	25	2	0	0
40	165	32	3	0	0
50	157	38	5	0	0
60	150	43	6	1	0
70	143	48	8	1	0
80	136	53	10	1	0
90	130	56	12	2	0
100	124	60	14	2	0

if a 60 m tower is located in a part of Florida where $N_g = 10 \text{ km}^{-2} \text{yr}^{-1}$ then the long-term average downward lightning incidence will be about 0.5 yr^{-1} (assuming $\alpha = 2$ and $\beta = 1$), that is, on average the tower will be struck every other year. The use of Eq. 2.6 would result in a lightning incidence value of about 1 yr^{-1} . For a house (or other similar structure) located in a region characterized by a moderate ground flash density of $4 \text{ km}^{-2} \text{yr}^{-1}$, and having an area of $10 \times 20 \text{ m}^2$ and a height of 5 m so that the equivalent attractive distance is about 10 m (extrapolating Eq. 2.9 with $\alpha = 2$ and $\beta = 1$ to a structure height of 5 m), the approximate equivalent attractive area is $30 \times 40 = 1200 \text{ m}^2$. Such a house is expected to be struck by lightning $1200 \text{ m}^2 \times 10^{-6} \text{ km}^2 \text{m}^{-2} \times 4 \text{ km}^{-2} \text{yr}^{-1} = 4.8 \times 10^{-3}$ times a year, or about once every 200 years. Another way to think of this lightning incidence is that, in this region, one in 200 houses will be struck each year, on average.

The probability that a structure, represented by its equivalent attractive area A , will be struck exactly 0, 1, 2, 3, ..., n times in T years can be estimated using the Poisson probability distribution,

$$p(n) = (Z^n/n!) \exp(-Z) \quad (2.10)$$

where $Z = AN_gT = N_dT$, the average number of strikes expected over T years, provided that N_g remains constant. Continuing the previous example, Table 2.4 gives the number of houses, out of a total of 200, expected to be struck by lightning n times over T years, as predicted by Eq. 2.10. Each number is found as the product of the total number of houses, 200, and $p(n)$ from Eq. 2.10, with subsequent rounding off to the nearest integer. In two cases ($n = 1$, $T = 80 \text{ yr}$ and $T = 100 \text{ yr}$) the rounded-off number was increased by unity in order to assure that the sum of numbers

in each row is 200, the total number of houses considered. It follows from Table 2.4 that, for example, over a 60-year period one should expect that 150 houses will not be struck at all, 50 ($= 43 + 6 + 1$) will receive at least one strike, six houses will be struck twice, and one house will be struck three times. If $N_g = 12 \text{ km}^{-2} \text{ yr}^{-1}$, which is characteristic of some areas in Florida, then over a period of 60 years (perhaps the lifetime of a house) only 84 houses out of the 200 will not receive any lightning strikes, and 11 houses will be struck three times or more.

Equation 2.10 was used by Eriksson and Meal (1984) and Anderson (1980) for calculating the number of power-line poles or towers not struck by lightning or struck exactly n times in six or eight years, respectively. The calculated results were found to be in good agreement with experimental data.

2.9.3. Upward flashes

Once the incidence of downward lightning N_d is found from Eq. 2.8 using the concept of an equivalent attractive area, the incidence of upward flashes N_u can be determined by subtracting N_d from N , the latter being found using Eq. 2.6. Recall that if the structure height is less than 100 m or so, it is usually assumed that $N_u = 0$. If only the percentage of upward flashes is sought, Eq. 2.7 can be used.

2.10. Summary

The global lightning flash rate is some tens to a hundred per second or so. The maximum flash rate in different storms varies typically from a few to some tens per minute, and the average flash rate per overall storm is about an order of magnitude lower. The majority of lightning discharges do not involve ground, the ratio z of cloud to cloud-to-ground flashes varying over a wide range, from about 2 to 10 or more. The most common measures of lightning incidence to an area are the annual number of thunderstorm days T_D , the annual number of thunderstorm hours T_H , and the annual ground flash density N_g . For a global flash rate of 100 s^{-1} and a global ratio z of cloud to cloud-to-ground flashes equal to 3, the average flash density over the entire Earth's surface is about $6 \text{ km}^{-2} \text{ yr}^{-1}$, and the ground (land or sea) flash density is about $1.5 \text{ km}^{-2} \text{ yr}^{-1}$. The annual ground flash density has been estimated from records of lightning flash counters and lightning locating systems. The observed variation in ground flash density from one region to another in the United States is more than two orders of magnitude. Many flashes strike ground at more than one point. When only one location per flash is recorded, the correction factor for measured values of ground flash density needed in order to take into account multiple channel terminations on ground is about 1.7. Lightning activity varies from year to year, and it is possible that the variation

of global lightning activity can be used for studying various aspects of global climate change, such as global temperature change. Ground-based objects with heights ranging from about 100 to 500 m experience both downward and upward flashes, the proportion being a function of the height of the object. Structures having heights less than 100 m or so are often assumed to be struck only by downward lightning, while those with heights greater than 500 m or so can be assumed to experience only upward flashes. A house located in a region characterized by a moderate ground flash density of $4 \text{ km}^{-2} \text{ yr}^{-1}$ and having an area of $10 \times 20 \text{ m}^2$ and a height of 5 m is expected to be struck by lightning roughly once every 200 years.

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